

Gliogenesis from the subventricular zone modulates the extracellular matrix at the glial scar after brain ischemia

Reviewed Preprint

v3 • January 20, 2025

Revised by authors

Reviewed Preprint

v2 • October 14, 2024

Reviewed Preprint

v1 • March 13, 2024

Maria Ardaya, Marie-Catherine Tiveron, Harold Cremer, Benjamin Dehay, Fernando Pérez-Cerdá, Carlos Matute, Federico N Soria , Fabio Cavaliere 

Achucarro Basque Center for Neuroscience, Leioa, Spain • Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain • CIBERNED, Madrid, Spain • Aix Marseille Univ, CNRS, IBDM, Marseille, France • Univ. Bordeaux, CNRS, IMN, UMR 5293, Bordeaux, France • Ikerbasque, Basque Foundation for Science, Bilbao, Spain • Basque Biomedical Platform for Human Research (BBioH), Leioa, Spain

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

The authors show that a middle carotid artery occlusion (MCAO) hypoxia lesion leads to hyaluronan-mediated chemoattraction to the lesion penumbra of Thbs4-expressing astrocytes of the sub-ventricular zone (SVZ). These findings are **valuable** because they shed light on the function of astrocytes from the adult SVZ in pathological states like brain ischemic injury. The results are **convincing**, as they rely on a comprehensive analysis of experimental data.

<https://doi.org/10.7554/eLife.96076.3.sa3>

Abstract

Activation of the subventricular zone (SVZ) following cerebral ischemia is one of the brain's early responses to counteract neuron loss and minimize tissue damage. Impaired brain regions communicate with the SVZ through various chemotactic signals that promote cell migration and differentiation, primarily involving neural stem cells (NSC), neuroblasts, or glioblasts. However, the activation of gliogenesis and the role of newly formed astrocytes in the post-ischemic scenario remain subjects of debate. We have previously demonstrated that adenosine release after brain ischemia prompts the SVZ to generate new astrocytes. Here, we used transient brain ischemia in mice to identify the cellular origin of these astrocytes within the SVZ neurogenic niche and to investigate their role in the pathological process. By combining immunofluorescence, BrdU-tracing, and genetic cell labeling, we tracked the migration of newborn astrocytes, positive for the proteoglycan marker Thbs4, from the dorsal and medial SVZ to the perilesional barrier surrounding the ischemic core, known as the “glial scar”. We found that these Thbs4-positive astrocytes modulate the dense extracellular matrix at the lesion border by both synthesizing and degrading hyaluronan. We also show that while the accumulation of hyaluronan at the lesion site is sufficient to recruit newborn astrocytes, its degradation at the SVZ correlates with gliogenesis. These findings suggest that newborn

astrocytes could be a promising pharmacological target for modulating the glial scar after brain ischemia and facilitate tissue regeneration.

Introduction

Cell regeneration in the adult mammalian brain primarily occurs in two neurogenic niches: the subventricular zone (SVZ) in the lateral ventricle and the subgranular zone (SGZ) in the dentate gyrus of the hippocampus (Bond et al., 2015). Adult brain stem cells are multipotent and originate from the embryonic radial glia, thus sharing key cell markers and features with resident astrocytes. In the adult mouse brain, neuroblasts from SVZ migrate through the rostral migratory stream (RMS) to the olfactory bulb (OB), where neurogenesis takes place (Luskin, 1993). Neuronal hyperactivity (e.g., during epilepsy), as well as pathological conditions (e.g., stroke) or natural (e.g., aging) processes, can impair the physiological functions of the neurogenic niches, altering their neurogenic and gliogenic potential (Chaker et al., 2016; Kralic et al., 2005; Sundholm-Peters et al., 2005). For instance, it is known that SVZ cells are activated *in vivo* after focal ischemia following a middle cerebral artery occlusion (MCAO) (Ceanga et al., 2021).

We have previously demonstrated in oxygen and glucose-deprived (OGD) organotypic brain slices that the ischemic region and the SVZ establish a biochemical interplay based on a gradient of toxic and protective factors released early after the insult (Cavaliere et al., 2006). Furthermore, others have demonstrated that brain stroke models in mice can activate astrogliogenesis at the expense of neurogenesis (Benner et al., 2013; Laug et al., 2019; Li et al., 2010). Benner and colleagues identified a SVZ-derived astrocyte population characterized by high levels of the membrane-bound proteoglycan thrombospondin 4 (Thbs4), a cell-cell and cell-matrix adhesion molecule that has been considered neuroprotective against ischemic damage (Benner et al., 2013; Laug et al., 2019). Ablation of Thbs4 in KO mice also produced abnormal ischemic scar formation and increased microvascular hemorrhage, suggesting a role of SVZ astrogliogenesis in brain damage and glial scar formation induced by ischemic stroke (Benner et al., 2013).

The perilesional barrier known as the *glial scar* assembles early after brain ischemia, isolating the damaged area and limiting both the spread of pathogens and potentially toxic damage-associated molecular patterns (DAMPs) into the surrounding area (Anderson et al., 2016; Bush et al., 1999; Faulkner et al., 2004). Consisting of packed reactive astrocytes and a dense extracellular matrix composed mostly of hyaluronic acid (HA), the glial scar represents a physical barrier for the SVZ-generated newborn neurons that migrate into the ischemic core (Arvidsson et al., 2002; Yamashita et al., 2006). Thus, modulating glial scar formation after brain ischemia may be a strategy to reduce tissue damage and restore brain functionality. In ischemic stroke, the brain extracellular matrix (ECM) goes through a significant reorganization due to injury and must undergo successful remodeling to promote brain repair. ECM elements involved in the ischemic cascade induce activation of astrocytes and microglia, which modulate ECM structure by releasing glycoproteins into the extracellular space (Adams and Gallo, 2018; Fawcett, 2015). This crosstalk culminates in the formation of the glial scar, which is created locally by reactive astrocytes and astrocytes-secreted ECM molecules such as HA and chondroitin sulphate proteoglycans (Bush et al., 1999; Lau et al., 2013).

We have previously demonstrated that high extracellular adenosine levels, a hallmark of ischemic insults, induce astrogliogenesis from the SVZ (Benito-Muñoz et al., 2016). Following our previous results, here we characterize the SVZ response to brain ischemia and investigate the role of newly generated astrocytes in the modulation of the glial scar after MCAO. We show that ischemic conditions increase the number of newly generated astrocytes in the SVZ and their migration to the ischemic area. We also propose the role of these astrocytes in modulating the extracellular matrix at the glial scar and suggest this astrocyte population as a possible target for brain repair after brain ischemia.

Results

Thbs4 is expressed in the neurogenic niche of the SVZ

As suggested by others (Benner et al., 2013 [↗](#); Girard et al., 2014 [↗](#); Pous et al., 2020 [↗](#)), Thbs4 is highly expressed in astrocytes of the SVZ and RMS and might play a role in ischemia-induced neurogenesis. To confirm these observations, we labeled the mouse SVZ by immunofluorescence for Thbs4. Thbs4 was mainly expressed in the SVZ, RMS, and OB but also in the cerebellum and ventral tegmental area (Fig. 1a [↗](#)). In the SVZ, the staining was associated with cells with different morphologies (Fig. 1b [↗](#)), ranging from cells with small primary apical processes in contact with the ventricle, to rounded NSC-like cells in the dorsolateral SVZ horn. After quantification by direct count of Thbs4-positive cells, Thbs4-positive astrocytes represented 21.4% of the total SVZ cells (Fig. 1c [↗](#)), with more than 60% of them displaying a radial glia-like type-B morphology (Bond et al., 2015 [↗](#); Doetsch et al., 1999 [↗](#)) (Fig. 1d [↗](#)). This is in accordance with other studies reporting Thbs4 expression in B-cells, either by immunohistochemistry (Beckervordersandforth et al., 2010 [↗](#); Benner et al., 2013 [↗](#)) or single-cell RNA sequencing (Basak et al., 2018 [↗](#); Cebrian-Silla et al., 2021 [↗](#); Llorens-Bobadilla et al., 2015 [↗](#)).

To validate the immunohistochemistry protocol for Thbs4, we labeled the SVZ cells with green fluorescent protein (GFP) under the control of the Thbs4 promoter (pThbs4-eGFP) by postnatal (P1) electroporation (Fig. 1e [↗](#)). Twenty-four hours after electroporation, almost 20% of total cells in the dorsal SVZ expressed GFP (Fig. 1f [↗](#)), with a greater expression in the rostral SVZ (Fig. 1g [↗](#)), confirming that Thbs4-positive cells were mainly localized in the neurogenic niche of the SVZ. This also suggests that Thbs4 is expressed in NSCs at early postnatal stages, generating astrocytes during development.

Thbs4 expression in the SVZ increases after brain ischemia

To investigate whether brain ischemia could activate astrogliogenesis in the SVZ, we analyzed cell proliferation and Thbs4 expression in the SVZ after MCAO. Sixty minutes of MCAO (Fig. S1a [↗](#)) caused neuronal damage in the cortex and striatum, as observed by cresyl violet staining (Fig. S1b [↗](#)) and NeuN immunofluorescence (Fig. S1c [↗](#)). The survival rate at 28 days post-lesion was around 50% (Fig. S1d [↗](#)). Moreover, motor deficits and animal weight of ischemic mice were monitored every day after MCAO for 30 days. Motor deficits improved overtime, while body weight dropped sharply the first three days after MCAO and was partially restored at 30 dpi (Fig. S1e, f [↗](#)).

To study cell proliferation in the SVZ, we administered 50 mg/kg BrdU i.p. every two hours (three doses in total) the day before MCAO (Fig. S2a [↗](#)). We observed an increase in the number of total SVZ BrdU-positive cells 24h after MCAO (Fig. S2b, c [↗](#)). Ki67 immunofluorescence of SVZ wholemount preparations (Fig. S2d [↗](#)) also revealed a more than tenfold increase of proliferative cells 24h after MCAO (Fig. S2e [↗](#)). In addition, we observed a significant increase in cleaved Caspase3-positive cells 24h post-lesion (Fig. S2f, g [↗](#)). These results suggest a fast and specific proliferative response in the ischemia-induced SVZ.

We next analyzed Thbs4 staining in combination with Nestin, an NSC marker, and Doublecortin (DCX), a marker of neuroblasts, before and after MCAO. The entire SVZ was analyzed in sham and ischemic mice 7-, 15- and 30-days post-ischemia (dpi). The immunofluorescence analysis of the SVZ (Fig. 2a [↗](#)) showed a transient decrease in Nestin and DCX-positive cells but an increase of Thbs4-positive astrocytes up to 15 dpi (Fig. 2b-d [↗](#)) compared to sham animals. The increase of Thbs4 astrocytes was also confirmed by Western blot analysis using SVZ homogenates (Fig. 2e, f [↗](#)), evidencing a 2-fold increased expression of Thbs4 in the ischemic SVZ. Similarly to immunofluorescence, Western blot also showed decreased Nestin protein levels at 15 dpi (Fig.

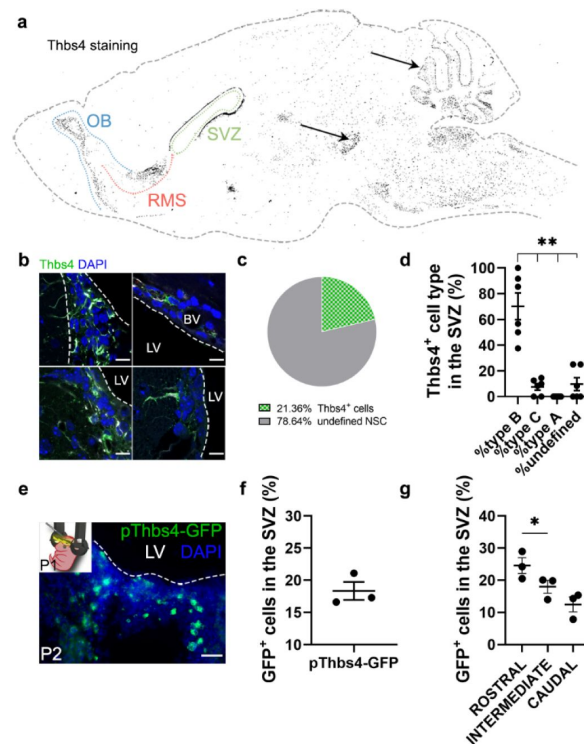


Figure 1

- Thbs4 labels type-B cells in the SVZ neurogenic niche

(a) Schematic drawing of the brain showing Thbs4 labeling throughout the mouse brain. Thbs4 was primarily found in the SVZ and RMS of the adult mouse telencephalon. Arrows indicate non-specific Thbs4 staining in cerebellum and ventral tegmental area. (b) Thbs4 positive cells in the SVZ displayed various morphologies. (c) Proportion of Thbs4-positive cells relative to the total NSCs population in the SVZ. (d) Classification of Thbs4-positive cells in the SVZ based to their NSC morphology. (e) Electroporation of pThbs4-GFP at P1 revealed Thbs4 expression in the postnatal dorsal SVZ. (f) Quantification of Thbs4-reporter expression showing the proportion of NSC expressing Thbs4 postnatally in the dorsal SVZ. (g) Thbs4-GFP reporter expression was upregulated in the rostral SVZ. BV: Blood vessel; LV: Lateral ventricle. Scale bar = 10 μ m (b) and 50 μ m (e). Bars represent mean $\pm$ SEM. * p < 0.05 and ** p < 0.01, Tukey post hoc test (after one-way ANOVA was significant at p < 0.05).

2g [↗](#)). These results demonstrate that Thbs4 expression in the SVZ increases after ischemia, coinciding with a decrease in neuroblast markers, suggesting that the classical pro-neuroblast program of the SVZ is interrupted in favor of the generation of astrocytes after ischemia.

Characterization of ischemia-induced cell populations in the SVZ

To characterize the different cell populations of the SVZ activated after focal ischemia, we labeled the whole pool of proliferating cells with 1% BrdU in drinking water for 14 days before MCAO (Codega et al., 2014 [↗](#)). One month after MCAO, BrdU in the SVZ primarily stained slow proliferative cells like type B cells, whereas in rapid transit-amplifying cells (type C), BrdU was diluted away. Therefore, we chose to label type C cells by intraperitoneal administration (50 mg/kg) of 5-Iodo-2'-deoxyuridine (IdU) 24h before euthanasia (**Fig. 3a** [↗](#)). To identify the cellular origin of newborn astrocytes, we analyzed the SVZ 30 dpi by immunofluorescence co-labeling Thbs4 and GFAP together with BrdU or IdU. We found a 2-fold increase in the total amount of proliferative cells (BrdU-positive) after MCAO with respect to the sham group (**Fig. 3b** [↗](#)), whereas we did not observe a significant change in the number of IdU-positive cells (**Fig. 3c** [↗](#)). The number of slow proliferative NSC (Thbs4/GFAP/BrdU) increased linearly after MCAO (**Fig. 3d, e** [↗](#)) especially in the dorsal region of the SVZ (**Fig. S3a** [↗](#)), whereas the number of activated NSC (Thbs4/GFAP/IdU) did not change significantly, either in the whole SVZ (**Fig. 3f** [↗](#)) or in any subregion analyzed (**Fig. S3b** [↗](#)). These results suggest that, after ischemia, Thbs4-positive astrocytes derive from the slow proliferative type B cells.

Ischemia-induced newborn astrocytes migrate from the dorsal SVZ to ischemic regions

To confirm that Thbs4-positive astrocytes originate from type B cells of dorsal SVZ, we labeled NSCs of dorsal SVZ by electroporating the pCAGGSx-CRE plasmid in P1 Ai14 Rosa26-CAG-tdTomato transgenic mice (**Fig. 4a** [↗](#)). Plasmid electroporation induced the expression of tdTomato (tdTOM) by Cre recombination, enabling the spatial tracking and cell fate assessment of electroporated NSCs from the dorsal SVZ. Three to four months after electroporation, we performed the MCAO, and tdTOM fluorescence was analyzed at 7, 30, and 60 dpi. To confirm the efficient labeling of NSCs by the electroporation protocol, we followed the tdTOM fluorescence in the OB (**Fig. S4a** [↗](#), see methods). Only those animals with tdTOM-positive cells in the OB were taken into account for later analysis. After quantification of tdTOM-positive NSCs 7, 30, and 60 dpi, we observed a significant decrease in tdTOM-positive cells in ischemic animals in all SVZ regions analyzed (**Fig. 4b, c** [↗](#) and **Fig S4b** [↗](#)), suggesting that they either left the area, or died. Conversely, there was a general increase of tdTOM-positive cells outside the SVZ (**Fig. 4d, e** [↗](#)), especially in the intermediate and caudal axis of the whole brain compared to sham animals (**Fig. S4d-g** [↗](#)), suggesting that electroporated cells had migrated away from the SVZ.

To identify the proportion of NSCs that differentiated into Thbs4-positive astrocytes following ischemia, we analyzed the co-expression of Thbs4 with tdTOM after MCAO. Despite the general decrease of tdTOM-positive cells in the SVZ, the fraction of Thbs4/tdTOM cells increased at 30 dpi after MCAO (**Fig. 4f, g** [↗](#)), reaching a maximum increase at 60 dpi (**Fig. 4h** [↗](#)), in particular in rostral and intermediate SVZ (**Fig. S4c** [↗](#)). Finally, we analyzed the Thbs4-positive astrocytes recruitment to the ischemic regions. More than half of the total tdTOM-positive cells in the infarcted area expressed the Thbs4 marker at 60 dpi (**Fig. 4i, j** [↗](#)). These results suggest that ischemia-induced astrogliogenesis in the SVZ occurs in type B cells from the dorsal region and that these newborn Thbs4-positive astrocytes migrate to the ischemic areas.

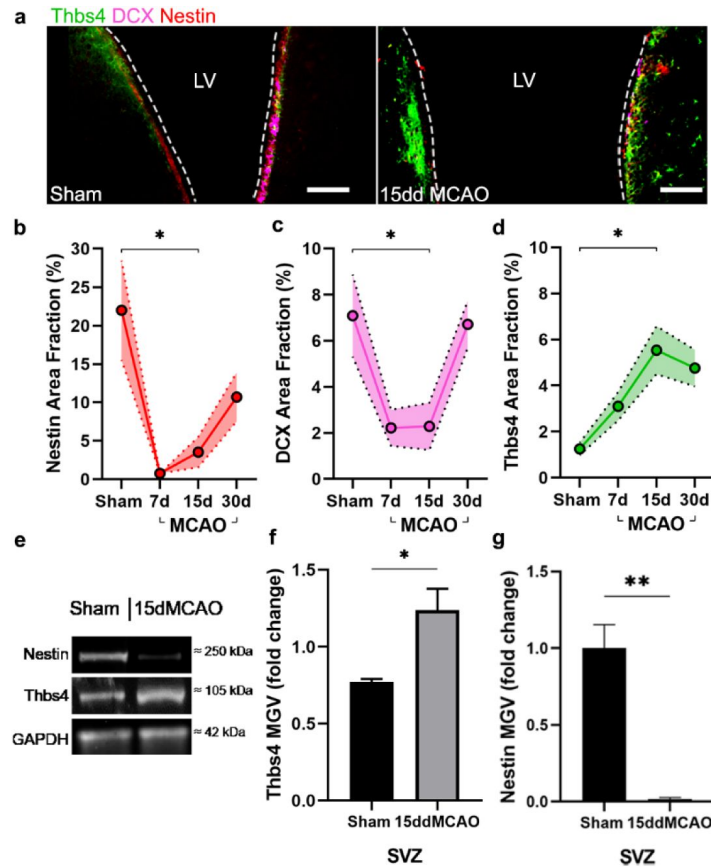


Figure 2

- Thbs4 expression is upregulated in the SVZ after brain ischemia

(a) Representative images of Thbs4 (green), Nestin (red) and DCX (magenta) in the SVZ under physiological conditions (left) and 15 days post-injury (dpi) (right). (b, c) The number of actively proliferating NSCs and neuroblast decreases at 7 and 15 dpi in the SVZ, as shown by reduced Nestin (b) and DCX (c) levels, respectively. (d) Thbs4 expression increases in the SVZ over time following ischemic injury. (e) Representative western blot images of Nestin and Thbs4 in sham and 15 dpi SVZ. (f, g) Quantification of mean gray value (MGV) from (e), normalized to GAPDH, shows an increased in Thbs4 (f) and a decrease in Nestin (g) in the 15 dpi SVZ. Scale bar = 100 μ m in (a). $n = 6$ (b, c, d) and 3 (e, f, g) per condition. Bars represent mean $\pm$ SEM. $*p < 0.05$ and $**p < 0.01$; two-tail Student's t test (sham vs. 15ddMCAO) and Tukey post-hoc test (after one-way ANOVA was significant at $p < 0.05$). See also Supplementary Figures 1 and 2.

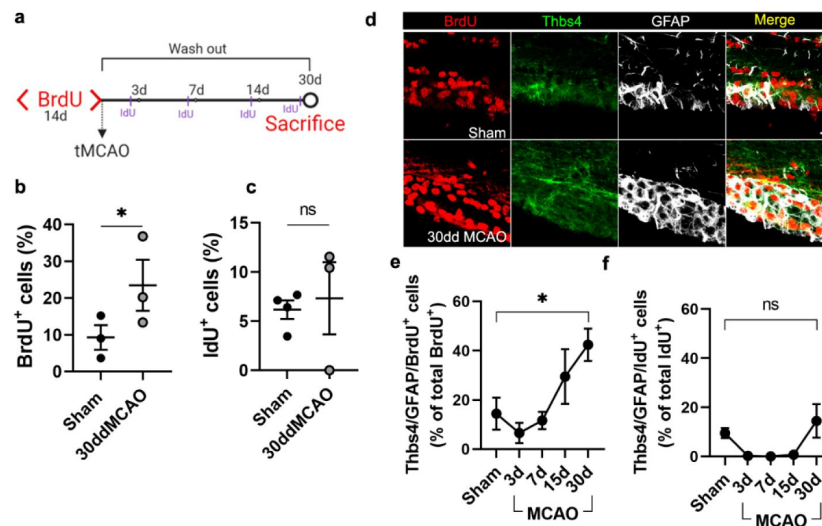


Figure 3

- Thbs4-positive cells derive from type-B NSCs in the SVZ after brain ischemia

(a) Experimental design: chronic BrdU (1% in water) treatment was administered for two weeks. BrdU labeling persisted only in B-type NSCs 30 days after treatment. Animals were injected with IdU (50 mg/kg) three times the day before euthanasia to label proliferating cells. (b, c) BrdU (b) and IdU (c) positive cells in the SVZ. Only BrdU-positive cells showed a significant increase at 30 dpi. (d) Representative confocal images showing Thbs4, GFAP and BrdU levels in sham (top) and 30 dpi mice (bottom). (e, f) Thbs4/GFAP/BrdU positive cells (slow proliferative cells) increased in the SVZ at 30 dpi (e), while no changes were observed in Thbs4/GFAP/IdU positive cells (fast proliferative cells). Scale bar = 10 μ m (d). n = 4 (sham) and 3 (MCAO). Bars represent mean $\pm$ SEM. *p < 0.05 by two-tail Student's t-test (sham vs. 30dd MCAO). See also Supplementary Figure 3.

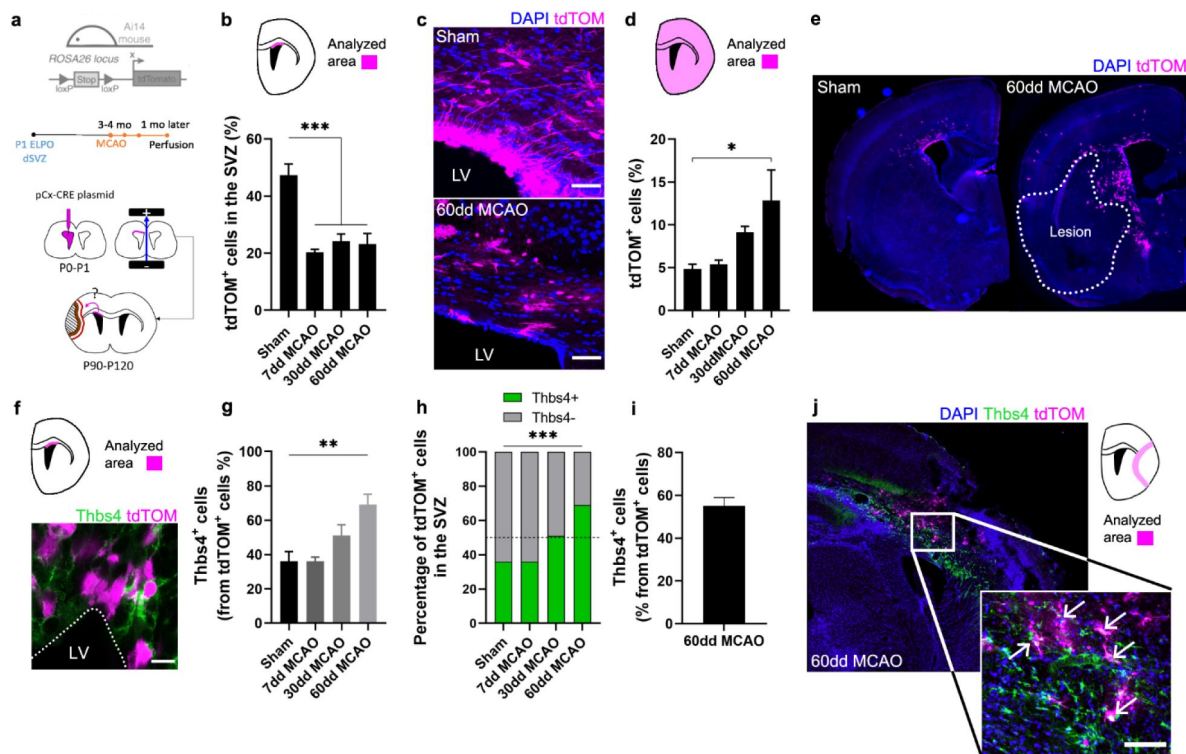


Figure 4

- Ischemia-induced Thbs4 astrocytes migrate from the SVZ to ischemic areas

(a) Experimental design: Ai14, ROSA26-CAG-tdTomato transgenic mice were electroporated in the dorsal SVZ at P1. The pCAGGSx-CRE plasmid induced tdTomato (tdTOM) expression in dorsal NSCs. MCAO was performed 3-4 months later in the same electroporated hemisphere. Mice were analyzed at 7, 30, and 60 dpi. (b) Quantification of tdTOM-positive cells shows a decrease in the dorsal SVZ following brain ischemia. (c) Representative images of tdTOM-positive cells in sham (top) and 60 dpi SVZ (bottom). (d) Quantification of tdTOM-positive cells in the whole brain shows a gradual increase outside the SVZ after brain ischemia. (e) Representative images of tdTOM expression in sham (left) and 60 dpi mice (right). (f) Representative image of Thbs4/tdTOM-positive cells in the dorsal SVZ. (g) Quantification shows an increase in Thbs4/tdTOM-positive cells in the SVZ at 30 and 60 dpi. (h) The proportion of Thbs4-positive cells within the tdTOM-positive pool increases in the SVZ at 30 and 60 dpi. (i) Around 50% of tdTOM-positive cells expressed Thbs4 in the infarcted tissue at 60 dpi. (j) Representative image of Thbs4 and tdTOM expression near the damaged area. Scale bar = 50 μ m (c), 10 μ m (f), and 100 μ m (g). $n = 6$ animals per condition. Bars represent mean $\pm$ SEM. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ by Chi-square test (h) and Tukey post-hoc test (after one-way ANOVA was significant at $p < 0.05$). See also Supplementary Figure 4.

Thbs4-positive astrocytes accumulate at the glial scar after brain ischemia

Besides its role in the neurogenic niches, Nestin is also rapidly and locally expressed by reactive astrocytes and participates in glial scar formation (Frisén et al., 1995 [\[1\]](#)). Thus, we used Nestin immunofluorescence as a reliable tool to identify the lesion border where the glial scar occurs (Fig. 5a [\[2\]](#)). Nestin staining of the scar was also compared with more classical methodologies like TTC staining (Fig. 5b [\[3\]](#)), where viable tissue is stained red and dead tissue is left unstained. The white-pale pink area was then measured as infarct tissue by TTC staining (Fig. 5c [\[4\]](#)). Unlike the TTC staining, which can only be performed shortly after the lesion, the visualization of the glial scar with Nestin immunofluorescence allowed us to analyze the lesion at later time-points. We observed a rapid formation of the glial scar as early as 7 dpi (Fig. 5d [\[5\]](#)). Analyzing the expression of the two astrocytic markers, Thbs4 and GFAP, as a percentage of the glial scar region-of-interest (ROI), we observed different dynamics and localization. GFAP expression increased rapidly after 7 days within the glial scar ROI, whereas Thbs4 was upregulated at 15 dpi (Fig. 5e [\[6\]](#)). This delay suggested that the glial scar harbors two different astrocytic sub-populations: GFAP-only or GFAP/Thbs4. To analyze the spatial location of GFAP and Thbs4 markers within the Nestin border, we measured the fluorescence intensity profile of damaged area at 30 dpi. GFAP was found in both the scar and core areas, whereas Thbs4 fluorescence was found mostly at the borders of the glial scar (Fig. 5f, g [\[7\]](#)). Analysis and dynamics of Thbs4-positive astrocyte migration were further performed in the corpus callosum (CC), striatum (Str), cortex (Ctx) and olfactory bulb (OB) (Fig. 55a [\[8\]](#)). We observed a linear increase of Thbs4-positive astrocytes from 7 to 60 dpi, both in the Ctx and Str (Fig. 55b [\[9\]](#)). Migration occurred mainly through the corpus callosum; among the total GFAP-positive astrocytes, 36.5% were also positive for Thbs4, suggesting that more than one-third of the astrocytes in the glial scar originated from the SVZ (Fig. 55c [\[10\]](#)). Likewise, the number of DCX-positive cells dramatically decreased 15 dpi in all OB layer (Fig. 55d-f [\[11\]](#)) together with a significant decrease of BrdU-positive cells (Fig. 55g [\[12\]](#)) DCX/BrdU-positive neuroblasts in the RMS (Fig. 55h [\[13\]](#)), suggesting that ischemia induced a change in the neuroblasts' ectopic migratory pathway, depriving the OB layers of SVZ newborn neurons.

Thbs4-positive astrocytes modulate extracellular matrix homeostasis at the glial scar

To investigate the role of Thbs4-positive astrocytes in the glial scar, we analyzed hyaluronic acid (HA), the main component of the brain extracellular matrix (ECM), using the hyaluronic acid binding protein (HABP) as a marker. Following ischemic damage, local astrocytes (GFAP+/Thbs4-) react by adopting a fibrotic morphology, producing high-molecular weight HA (HMW-HA) to form the glial scar and isolate the damaged area (Preston and Sherman, 2011 [\[14\]](#)). We then examined HA deposition in the corpus callosum and the infarcted areas after MCAO. HABP immunofluorescence (Fig. 6a [\[15\]](#)) and image analysis revealed HA accumulation in the infarcted areas. Skeleton analysis, which quantifies the length of HA chains and correlates with its molecular weight, identified progressively longer HA cable-like structures at the glial scar as the infarct evolved (Fig. 6b [\[16\]](#)). Fractal dimension analysis, which reflects the interconnectivity of the extracellular matrix (Soria et al., 2020 [\[17\]](#)), further confirmed HA accumulation and increased matrix complexity in infarcted areas, but not in corpus callosum (Fig. 6c [\[18\]](#)), suggesting that ECM accumulation is different in white and gray matter.

The proximity of Thbs4-positive astrocytes to HA in the ischemic scar (Fig. 6d, e [\[19\]](#)) led us to investigate if this population contributed to ECM production and, consequently, to scar formation. Since HA is produced exclusively at the cell membrane (Sherman et al., 2015 [\[20\]](#); Zimmermann and Dours-Zimmermann, 2008 [\[21\]](#)), we analyzed HA labeling within 1 μ m of the Thbs4-positive cell

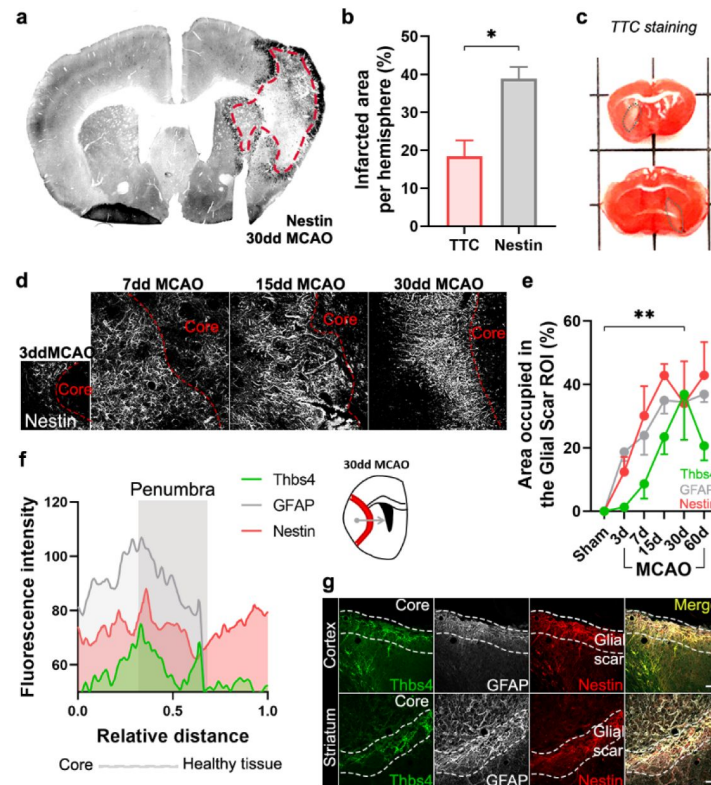


Figure 5

- Ischemia-induced Thbs4 astrocytes accumulate at the glial scar

(a) The glial scar could be observed at 30 dpi using Nestin immunofluorescence. (b) Classical TTC technique underestimated ischemic regions compared with Nestin immunofluorescence. (c) Representative images of TTC staining in infarcted brains. (d) Representative confocal images showing the time course of Nestin expression at 3, 7, 15 and 30 dpi. (e) The Nestin-positive glial scar was established by 15 dpi, while Thbs4 astrocytes arrived at the glial scar later, by 30 dpi. (f) Fluorescence profile of Thbs4, GFAP and Nestin marker from the core to penumbra areas shown Thbs4 was mainly located at the borders of the Nestin-positive glial scar. (g) Representative images of Thbs4 (green), GFAP (white) and Nestin (red) markers in sham (top) and 30 dpi mice (bottom). Scale bar = 50 μ m (g). n = 3 animals per condition. Bars/lines represent mean $\pm$ SEM. * p < 0.05 and ** p < 0.01 by two-tail Student's t test (TTC vs. Nestin) and by Dunnett posthoc test (after two-way ANOVA was significant at p < 0.05). See also Supplementary Figure 5.

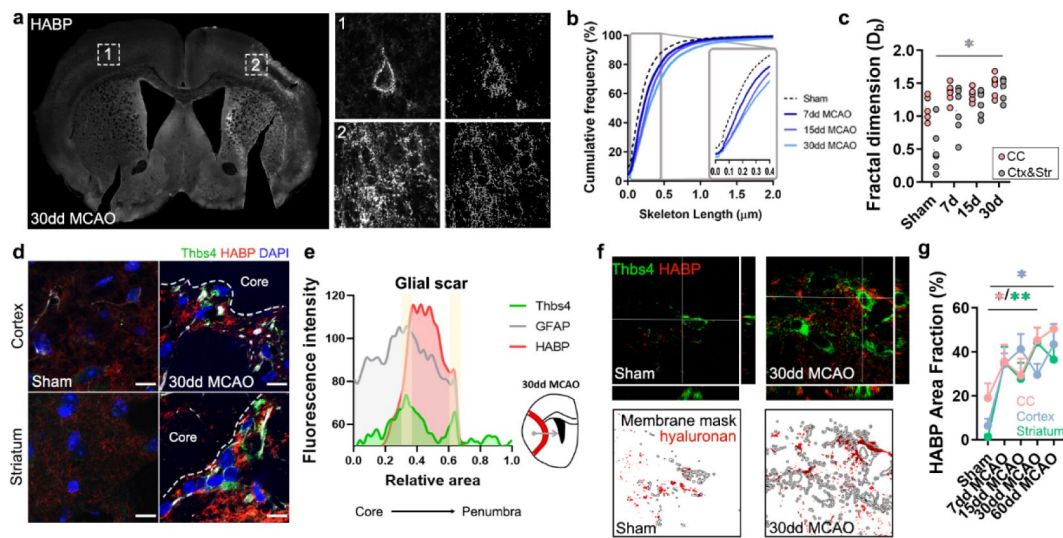


Figure 6

- Thbs4-positive astrocytes produce hyaluronan at the glial scar after MCAO

(a) Representative images of hyaluronan (HA) labeled with HABP in the contralateral (1) and ipsilateral hemisphere (2). Only perineuronal nets are clearly visible in the healthy tissue, while a dense interstitial matrix is observed in ischemic areas. (b) Cumulative distribution of skeletonized HA signal in sham and infarcted regions of 7, 15 and 30 dpi mice ($p < 0.0001$). (c) Quantification of fractal dimension showed increased extracellular matrix interconnectivity in the infarcted cortex and striatum (Ctx & Str) at 30 dpi. (d) Representative confocal images of Thbs4 (green) and HABP (red) in the cortex and striatum of sham and 30 dpi mice. Note the HA labelling on the surface of Thbs4-positive cells. (e) Fluorescence intensity profile of Thbs4, GFAP and HABP markers from the lesion core to penumbra areas. Thbs4 expression was mainly restricted to the borders of the HA-positive glial scar at 30 dpi. (f) Representative confocal images of Thbs4 and HABP markers in sham and 30 dpi mice (top). Examples of HA (red) occupying the membrane mask of Thbs4-positive astrocytes in sham and 30 dpi mice (bottom), showing increased HA synthesis after MCAO. (g) Quantification of membrane-bound HA showed increased HA along Thbs4-positive membrane after brain ischemia (corpus callosum: pink; infarcted cortex: blue; infarcted striatum: green). Scale bar = 10 μm (e). $n = 6$ animals per condition. $*p < 0.05$ and $***p < 0.0001$ by Kruskal Wallis test and two-way ANOVA (significant at $p < 0.05$). See also Supplementary Figures 6 and 7.

membrane in high-resolution confocal images (**Fig. 6f**). Image analysis showed a significant increase in HABP area at Thbs4-positive astrocyte-cell membranes after MCAO (**Fig. 6g**), suggesting a role of Thbs4-positive astrocytes in HA production within the glial scar.

To explore the potential role of Thbs4-positive astrocytes in HA removal, we quantified intracellular HABP spots in Thbs4-positive astrocytes at the glial scar. While HMW-HA cleavage occurs extracellularly, HA fragments can be internalized for degradation by lysosomal hyaluronidases (Preston and Sherman, 2011; Zimmermann and Dours-Zimmermann, 2008). Intracellular HABP spots were compared between Thbs4-positive astrocytes and local astrocytes (GFAP-positive/Thbs4-negative) (**Fig. S6a**). Thirty days after MCAO, Thbs4-positive astrocytes showed more intracellular HABP than local astrocytes in the infarcted cortex (**Fig. S6b**). However, the number of HA spots internalized was higher in local astrocytes compared with Thbs4 astrocytes (**Fig. S6c**), suggesting that Thbs4-positive cells internalize HA in fewer vesicles. To confirm that astrocytes were degrading HA, we performed an immunofluorescence analysis of CD44 (**Fig. S6d**). CD44 is a transmembrane glycoprotein receptor that mediates HA internalization and degradation, and therefore ECM remodeling (Dzwonek and Wilczynski, 2015). We detected that around 50% of the total Thbs4-positive astrocytes contained internalized CD44 receptor, in the striatum 30 dpi (**Fig. S6e**), with higher numbers (60%) for the infarcted corpus callosum. These findings suggest that Thbs4-positive astrocytes participate in both the formation and degradation of the extracellular matrix at the glial scar.

To further investigate HA internalization by astrocytes, we conducted *in vitro* experiments using cultures of rat SVZ neural stem cells, a more versatile preparation that displays the same HA internalization and degradation machinery as mouse brain tissue (Sherman et al., 2015). An oxygen and glucose deprivation (OGD) protocol was applied as an ischemia model and cells were fixed 7 days after OGD (**Fig. S7a**). HA was analyzed via HABP immunofluorescence (**Fig. S7b**), both inside and outside the Thbs4-positive cells (**Fig. S7c**). Thbs4-positive cells showed increased intracellular HA after OGD (**Fig. S7d**), while extracellular HA levels decreased (**Fig. S7e**). However, this phenomenon was reversed after adding hyaluronidases in the extracellular space (**Fig. S7f, g**), suggesting that Thbs4-positive cells from SVZ homogenates can internalize or synthesize HA depending on external stimuli. HA internalization was also measured in Iba1-positive cell cultures where we observed a reduced internalization compared to Thbs4-positive cells (**Fig. S7h**). Lastly, metalloproteinase activity was found to be increased in Thbs4 cultures following OGD (**Fig. S7i**). Altogether, these results suggest Thbs4-positive cells can internalize HA after ischemic conditions but also synthesize it when ECM is degraded.

Hyaluronan accumulation is sufficient but not exclusive for the recruitment of Thbs4-positive astrocytes

Since Thbs4-positive astrocytes migrate to the glial scar but not to the ischemic core, we hypothesized that HA, the primary component of the ischemic scar, may be the trigger signal for recruiting Thbs4-positive astrocytes. To test this hypothesis, we simulated the ECM accumulation that occurs at the scar by overexpressing the Hyaluronan synthase 2 (HAS2) from the naked mole rat (*Heterocephalus glaber*), a tool often used in mice to produce HA of very high molecular weight (Tian et al., 2013; Zhang et al., 2023). We used a viral vector (AAV2/9-CAG-HAS2-3xFlag) with a FLAG tag to visualize transduced cells after intracerebral inoculation in the striatum and checked HA accumulation in FLAG-positive areas using HABP staining (**Fig. 7a**). We overexpressed HAS2 (or GFP as control) unilaterally. In a subset of animals that underwent MCAO, AAV-HAS2 was injected contralateral to the infarct, to compare Thbs4-positive astrocytes recruitment by both stimuli, the inflammation from the ischemic damage and the HA accumulation *per se* (**Fig. 7b**). FLAG-positive cells were quantified as a surrogate marker of viral infection for both control and MCAO groups (**Fig. 7c**). The GFP control group did not increase HA production *per se* (**Fig. 7d**). However, HA overexpression was observed in HAS2 groups (both control and MCAO group) by HABP immunofluorescence (**Fig. 7e, f**).

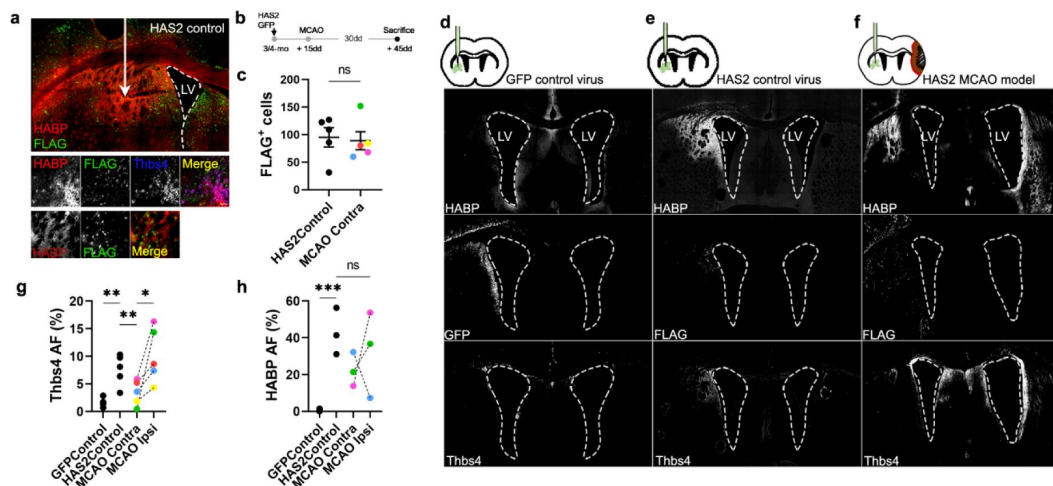


Figure 7

- Hyaluronan accumulation is sufficient but not exclusive to recruit Thbs4 astrocytes

(a) Representative images of HA accumulation caused by viral-mediated overexpression of HAS2 from naked mole rat. (b) Experimental design: HAS2 viral injections were performed 2 weeks before MCAO. Animals were sacrificed 30 days after MCAO. We used standard GFP virus as control. Two groups of HAS2-injected mice were studied, with only one subjected to MCAO. (c) FLAG-positive cells (HAS2 virus tag) did not show changes in either HAS2 control (HAS2Control) or HAS2-MCAO group (contralateral hemisphere). (d, e, f) Representative images of HABP, GFP and Thbs4 markers in GFP control virus (d), HAS2 control virus (e) and HAS2 MCAO group (f). (g) Thbs4 expression increased after HAS2 viral-mediated overexpression. However, Thbs4 was upregulated in the ischemic hemisphere when MCAO was performed together with the HAS2 viral infection (MCAO ipsi). GFP control virus did not induce increase of Thbs4. (h) HABP was used as a control for HA accumulation. Thbs4 astrocytes in HAS2 MCAO group increased in the ischemic hemisphere even though HA accumulation in both hemispheres (HAS2 and MCAO hemisphere), suggesting additional ischemia-induced signals recruit Thbs4 astrocytes to infarcted areas. Scale bar = 100 μ m (a). $n = 6$ animals per condition. $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$ by two-tail Student's t-test (HAS2Control vs. MCAO Contra) and Tukey posthoc test (after one-way ANOVA was significant at $p < 0.05$).

HA accumulation was sufficient to recruit the Thbs4-positive astrocytes from the ipsilateral SVZ 45 days after AAV-HAS2 inoculation (**Fig. 7g**). with a similar time window to that observed after MCAO. Interestingly, when we induced a lesion by MCAO and performed viral overexpression of HAS2 in the contralateral hemisphere, we found that Thbs4-positive astrocytes preferentially migrated to the ischemic glial scar (**Fig. 7g**). This migration of Thbs4-positive astrocytes to the site of ischemic damage occurred despite no differences in HA overexpression between contralateral (HAS2 infected) and ipsilateral (infarcted area) hemispheres (**Fig. 7h**). These results suggest that the accumulation of HMW-HA alone is sufficient to activate and recruit Thbs4-positive astrocytes from the SVZ. However, the preferential migration to the ischemic scar indicates that brain ischemia generates a more potent biochemical signal that not only activates the SVZ but also recruits newborn astrocytes.

Ischemia-induced gliogenesis at the SVZ correlates with local hyaluronan degradation

It is known that the niche microenvironment has a unique ECM composition (Kjell et al., 2020) and that the stiffness of the niche plays a crucial role in the proliferation and differentiation of progenitor cells (Long and Huttner, 2019; Segel et al., 2019). After observing the close association between Thbs4-positive astrocytes and the ECM at the glial scar, we decided to study the Thbs4/HA tandem in the SVZ following MCAO (**Fig. 8a**). We did not observe any significant changes in the HA area across the entire SVZ but noted a decrease at 7 and 15 dpi in the dorsal SVZ (**Fig. 8b-d**). Skeleton analysis showed fragmentation of HA exclusively in the dorsal SVZ at 7 and 15 dpi (**Fig. 8e-g**), i.e., earlier than the migration of these newborn astrocytes toward the lesion.

Next, we examined the expression of genes related to HA degradation (*Hyase 1, 2 and 3*) and synthesis (*HAS 1 and HAS 2*). We also analyzed *Thbs4* and hypoxia-inducible factor 1-alpha (*Hif1a*), which are known to be upregulated in the SVZ following brain ischemia (Benito-Muñoz et al., 2016; Liu et al., 2007). Fresh SVZ tissue was collected from sham, 15 and 30 dpi mice (**Fig. 8h**). We observed an increased expression of all Hyases at 15 dpi and slightly decreased by 30 dpi, whereas synthases (HAS) showed a gradual increase from 15 to 30 dpi (**Fig. 8i**). Additionally, we observed an upregulation of *Thbs4* and *Hif1a* both at 15 and 30 dpi (**Fig. 8i**). These results suggest that HA degradation in the SVZ is correlated with the *Thbs4* response to ischemia.

Discussion

The effects of brain insult on adult neurogenesis have been extensively discussed, both in animal models and humans. However, whether these injuries can trigger the generation of newborn astrocytes from the neurogenic niches remains a topic of debate. Neural stem cells in the adult brain are a remnant of embryonic radial glia. Some researchers suggest that depletion of neurogenesis in the adult brain is a consequence of astrogliogenesis (Encinas et al., 2011), while others propose astrogliogenesis is part of a neurorestorative program originating from the neurogenic niches (Bonzano et al., 2018; Sohn et al., 2015). Despite these differing perspectives, direct and definitive evidence of the generation of newborn astrocytes from the SVZ or the SGZ has yet to be presented, as well as the potential role of these astrocytes in aging and pathological conditions. In this study, we confirm that a strong and acute damage-associated stimulus in the brain, such as an ischemic infarct, prompts astrogliogenesis in the SVZ. Furthermore, we demonstrate that these astrocytes, which are positive for *Thbs4*, migrate to the glial scar and play a role in its modulation, being capable of synthesis and degradation of hyaluronan, a key component of the ECM at the scar.

Our findings align with previous studies, where cortical injury initiated by photothrombotic/ischemia was shown to trigger a substantial production of *Thbs4*-positive astrocytes from the SVZ, a process modulated by Notch activation (Benner et al., 2013). In this

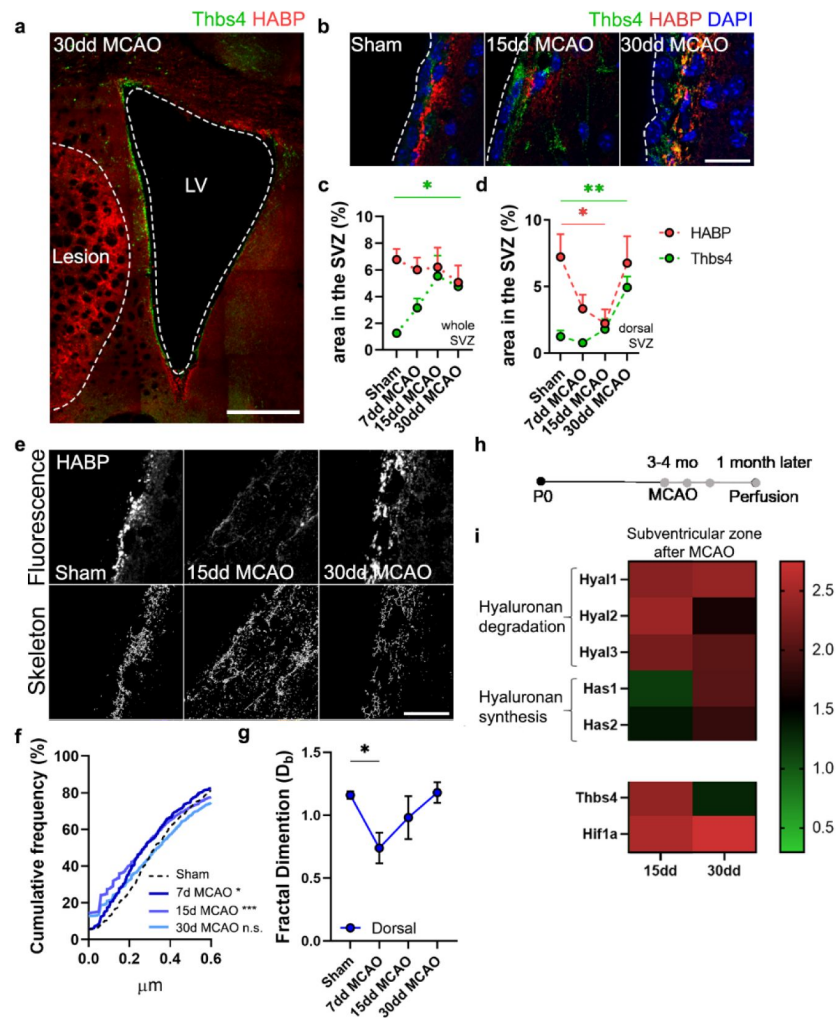


Figure 8

- HA degradation in the SVZ correlates with Thbs4 response to brain ischemia

(a) Representative image of Thbs4 and HA (HABP) markers in the SVZ at 30 dpi. (b) Representative images of Thbs4 and HA in the sham and 15, 30 dpi SVZ. (c, d) HA levels did not change in the entire SVZ after MCAO (c) but decreased significantly in the dorsal SVZ (d). Thbs4 expression increased throughout the SVZ after MCAO (c), with a marked rise at 30 dpi in the dorsal SVZ (d), coinciding with HABP reduction. (e) Representative confocal (top) and skeletonized (bottom) images of HABP in the dorsal SVZ of sham, 15 and 30 dpi mice. (f) Cumulative distribution of HA skeletons showed a reduction in the length of HA cable-like structures only in the dorsal SVZ at 7 and 15 dpi. (g) The interconnectivity of the extracellular matrix decreased by 7 dpi in the dorsal SVZ, as measured by fractal dimension. (h) Experimental design for qPCR: 3–4-month-old mice were subjected to MCAO, and fresh SVZ tissue was extracted 15 and 30 days after MCAO for RT-qPCR analysis. (i) Hyaluronan degradation genes (Hyal1, Hyal2 and Hyal3) increased by 15 and 30 dpi in the SVZ, while hyaluronan synthase genes (HAS1 and HAS2) were overexpressed later, at 30 dpi. Thbs4 and Hif1a were also upregulated by 15 and 30 dpi in the SVZ. Scale bar = 100 μ m (a). $n = 6$ (c, d, e and g) and 3 (i). $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$ by Kruskal Wallis test and Tukey post-hoc test (after one-way ANOVA was significant at $p < 0.05$).

work, the authors suggested a clear role for Thbs4-positive astrocytes in modulating the ischemic scar. In fact, homozygotic deletion of the Thbs4 gene led to abnormal glial scar formation and increased microvascular hemorrhage following brain ischemia. Similarly, our results demonstrate that the adult neurogenic niche at the SVZ responds early after ischemic stroke, degrading local hyaluronan plausibly to modify niche stiffness and facilitate proliferation and differentiation of NSCs. Further experiments using Thbs4-knockout models will help confirm this hypothesis.

By labeling NSC via *in vivo* postnatal electroporation, we provided proof of the origin and migration of newborn Thbs4-positive astrocytes from the SVZ to the ischemic scar in response to the injury. Notably, Thbs4 is expressed in the newborn astrocytes that migrate to the scar, whereas local reactive astrocytes at the site of injury express only GFAP. It is likely that SVZ-derived newborn astrocytes require Thbs4 expression to facilitate their migration to the ischemic area, as Thbs4 deletion has been shown to impair migration of SVZ-derived cells along the RMS (Girard et al., 2014 [↗](#)).

It is noteworthy that while the local, “GFAP-only”, reactive astrocytes at the scar are capable of forming the required ECM (Cregg et al., 2014 [↗](#); Wanner et al., 2013 [↗](#)), we show that Thbs4-positive (Thbs4/GFAP) astrocytes are also recruited from the SVZ to modulate scar formation. Unlike local reactive astrocytes, which are hastily generated and present both at the core and periphery of the lesion, the newborn astrocytes from the SVZ arrive later to the injury. We demonstrate that these Thbs4-positive astrocytes migrate from the SVZ to the lesion and halt at the lesion border (precisely defined by Nestin staining), establishing residence at the scar but not beyond.

The molecular signals that trigger gliogenesis in the SVZ and recruit newborn astrocytes to the lesion remain unclear. We previously identified adenosine, a byproduct of ATP hydrolysis, as a strong candidate, as it stimulates astrogliogenesis through A1 receptors following OGD *in vitro* and ischemia *in vivo* (Benito-Muñoz et al., 2016 [↗](#)). Here, we show that viral-mediated overproduction of HA, the primary component of the glial scar, is sufficient to activate the generation and migration of the Thbs4-positive astrocytes from the SVZ. Loss-of-function studies, using HA-depletion models or HA synthase (Has)-deficient mice are still needed to corroborate this finding, although the inflammation associated with HA deficiency might confound interpretation. Indeed, our results also show that the ischemic infarct, with its proinflammatory (cytokines, chemokines) and proteolytic (MMPs, ROS) signals (Jayaraj et al., 2019 [↗](#)), serves as a more potent attractor than the ECM alone.

The glial response from the SVZ we propose in this study might have several functional implications. Thbs4-positive astrocytes generated from the SVZ might act as a second wave of reactivity, supporting local astrocytes in modulating ECM formation. Hyper-reactive astrocytes at the lesion border decrease over time, along with reductions in HA synthesis and CD44 activity, suggesting that the second wave of Thbs4-positive astrocytes from the SVZ could sustain high levels of HA production (Lindwall et al., 2013 [↗](#)). Thus, Thbs4-positive astrocytes may be crucial during this delayed temporal window when glial scar remodeling is essential for neurological recovery (Cai et al., 2017 [↗](#); Dzyubenko et al., 2018 [↗](#)). However, it's important to note that only about one-third of the astrocytes at the glial scar are derived from the SVZ. This suggests that while SVZ-derived astrocytes may be important for glial scar maintenance, local astrocytes remain the central players in this process.

Based on our observations of both HA production and internalization by Thbs4-positive astrocytes *in vivo* within the ischemic region and *in vitro* following OGD, we propose a role for these astrocytes in the modulation, and possibly maintenance, of the ECM at the scar. It is likely that ECM degradation at the scar is also carried out by professional macrophages, such as microglia, which are known to be recruited to the scar (Fawcett and Asher, 1999 [↗](#)) and have been shown to degrade ECM through phagocytosis (Nguyen et al., 2020 [↗](#); Soria et al., 2020 [↗](#)). Understanding

which cells within the scar environment participate in its modulation, particularly its degradation, holds highly therapeutic value for promoting regeneration (Dzyubenko et al., 2018 [DOI](#); Silver and Miller, 2004 [DOI](#)). Here, we shed light on the unknown role of SVZ-derived astrocytes activated by ischemic brain injury and propose them as critical players in tissue regeneration after brain ischemia.

Methods

Animals

Male and female wild-type C57BL/6/J mice and Ai14-Rosa26-CAG-tdTomato^{flx/flx} transgenic mice bred at the animal facility of UPV/EHU were used for *in vivo* experiments. Sprague Dawley rat pups from P4 to P7 were used for *in vitro* experiments. All animals for *in vitro* and *in vivo* studies were handled in accordance with the European Communities Council Directive (2010/63/EU). All possible efforts were made to minimize animal suffering and the number of animals used. Animals were maintained under standard laboratory conditions with food and water *ad libitum*.

Transient MCAO

Brain ischemia was induced in 3–4-month old C57BL/6 mice (approximately 25 g) by occluding 60 minutes of the lenticulo-striate arteries with a 10 mm length of silicone-coated 6-0 monofilament nylon suture (602223PK10, Doccol Corporation) using a modification of the protocol described by (Gelderblom et al., 2009 [DOI](#)). Arteries occlusion generated a specific lesion in the cortex and striatum that was evaluated at different times after the MCAO (7, 15, 30 or 60 days after occlusion). Ischemic damage was evaluated in the ipsilateral hemisphere by Cresyl violet staining, Nestin immunofluorescence or hyaluronan staining. Sham animals, in which arteries were visualized but not disturbed, were used as an experimental control. Both experimental groups, sham and ischemic, were anesthetized with a mix of isoflurane and oxygen (induction of anesthesia with 4% isoflurane and maintenance for surgery at 1.5% isoflurane) and maintained under analgesia with buprenorphine 24 hours after ischemia (0.03 mg/Kg body weight i.p. every 12h). Animal wellness was evaluated daily, controlling the body weight and the motor deficit of the animal by means of neurological scores and pole test. Motor deficit was assessed in each animal 1h after MCAO and later every 24h. Animals without neurological deficits were excluded from the study.

5-bromo-2'-deoxyuridine (BrdU) protocol

The total pool of proliferating cells was labelled with three injections of 50 mg/kg 5-bromo-2'-deoxyuridine (BrdU) the day before MCAO protocol. Animals were sacrificed the day after MCAO to analyse acute SVZ proliferation. Chronic BrdU protocol was performed for quiescent NSC identification (Codega et al., 2014 [DOI](#)). Briefly, 1% of BrdU was administered in the drinking water for 14 days. On day 15, BrdU was removed and animals were subjected to MCAO protocol. Mice were sacrificed and perfused at 3, 7, 15, and 30 days after MCAO, although only animals with a temporal window of 30 days between MCAO and perfusion were used for statistical analysis. Activated cells were labelled by i.p. injection of 50 mg/Kg 5-iodo-2'-deoxyuridine (IdU) 24 hours before euthanasia.

Electroporation

Ai14-Rosa26-CAG-tdTomato^{flx/flx} transgenic mice and pCAGGS-CRE plasmid were kindly provided for Prof. Harold Cremer and Marie-Catherine Tiveron from Institut de Biologie du Développement de Marseille (IBDM). Postnatal electroporation was performed as described previously (Platel et al., 2019 [DOI](#)). Mice pups from P0–P2 were anesthetized by hypothermia. pCAGGS-CRE plasmid was used at a final concentration of 2.5 µg/ul and 2 µl was injected in the lateral ventricle by expiratory pressure using a glass micropipette. Following injection, pup brains were placed between

electrodes targeted the dorsal SVZ and subjected to five 95V electrical pulses (50 ms, 950 ms intervals). BTX ECM399 (Thermo Fisher Scientific) electroporator was used with 7 mm tweezer electrodes. Electroporated pups were reanimated at 37°C before returning to the mother.

In order to analyze the Thbs4 positive cells in the SVZ, we subcloned the eGFP gene (from a pCMV-eGFP N1) into a PGL3-basic backbone containing mouse Thbs4 promoter, kindly provided by Stenina-Adognravi lab (Muppala et al., 2017 [↗](#)). pThbs4-GFP was also electroporated in P0-P2 mice, and euthanized 1-day post-electroporation to corroborate Thbs4 expression in the SVZ at postnatal stages.

AAV vector production

AAV2/9-CAG-HAS2-3xFlag vector was produced by polyethylenimine (PEI) mediated triple transfection of low passage HEK-293 T/17 cells (ATCC; cat number CRL-11268). The respective AAV expression plasmid expression plasmid pNMRHAS2_N1 (provided by Dr. Vera Gorbunova) was cotransfected with the adeno helper pAd Delta F6 plasmid (Penn Vector Core, cat # PL-F-PVADF6) and AAV Rep Cap pAAV2/9 plasmid (Penn Vector Core, cat # PL-TPV008). AAV vector was purified as previously described (Bourdenx et al., 2015 [↗](#)). Cells are harvested 72 h post-transfection, resuspended in lysis buffer (150 mM NaCl, 50 mM Tris-HCl pH 8.5) and lysed by 3 freeze-thaw cycles (37°C/-80°C). The cell lysate is treated with 150 units/ml Benzonase (Sigma, St Louis, MO) for 1 h at 37°C, and the crude lysate is clarified by centrifugation. Vectors are purified by iodixanol step gradient centrifugation and concentrated and buffer-exchanged into Lactated Ringer's solution (Baxter, Deerfield, IL) using vivaspin20 100 kDa cut-off concentrator (Sartorius Stedim, Goettingen, Germany). Titrations were performed at the transcriptome core facility (Neurocentre Magendie, INSERM U862, Bordeaux, France). The genome-containing particle (gcp) titer was determined at a concentration of 1.10^{13} gcp/ml by quantitative real-time PCR using the Light Cycler 480 SYBR green master mix (Roche, cat # 04887352001) with primers specific for the AAV2 ITRs (fwd 50-GGAACCCCTAGTGATG GAGTT-3'; rev 50-CGGCCTCAGTGAGCGA-30) on a Light Cycler 480 instrument. The purity assessment of vector stocks was estimated by loading 10 µl of vector stock on 10 % SDS acrylamide gels; total proteins were visualized using the Krypton Infrared Protein Stain according to the manufacturer's instructions (Life Technologies).

In vivo AAV-HAS2 injections

Mice were anesthetized with isoflurane and fixed to a stereotaxic frame connected with a gas mask. We injected 1 µl of adeno-associated virus (AAV2/9) CAG-HAS2-3xFlag (1×10^{13} gc/µl) in the contralateral striatum (AP: +1.0; ML: -1.8; DV: -3.2) to simulate scar tissue hyaluronan. After two weeks, the ischemic group was subjected to the MCAO model in the ipsilateral hemisphere and perfused 30 days later. Only slices with representative lesions in both hemispheres were used for image and statistical analysis.

Immunofluorescence

Mice were anesthetized with an intraperitoneal injection of pentobarbital (100 mg/kg). Brain tissues were fixed by intracardiac perfusion using 4% paraformaldehyde. Fixed brains were sliced at 40 µm of thickness using the Leica VT1200S vibratome (Leica Microsystems) and subjected to immunofluorescence. For standard immunofluorescence protocol, tissue slices were permeabilized with 0.1% Triton and non-specific epitopes blocked with 1% Bovine Serum Albumin and 10% Donkey serum in PBS. Primary antibodies (Supplementary Table 1) were incubated at different concentrations (see **Table 1** [↗](#) for specification) overnight at 4°C and then washed three times with 0.1% Triton in PBS. Secondary conjugated antibodies were incubated for 1 h at room temperature. After three washes with 0.1% Triton in PBS, cells were stained for 10 min with DAPI and further washed with PBS. Finally, coverslips were mounted with Fluoromount (SouthernBiotech) and analyzed by fluorescence using the confocal microscope Leica TCS SP8. For hyaluronic acid labeling, endogenous biotins were pre-blocked using Streptavidin/Biotin blocking

solution (Vector Labs, #SP-2002) following the manufacturer's instructions and further blocked with PBS 1X 0.1% Saponin 1% BSA for one hour at room temperature. Primary antibodies and biotinylated Hyaluronic Acid Binding Protein (HABP, #385911 Millipore) were incubated for 72 hours at 4°C or 24 hours at room temperature. After three washes with PBS, 1X 0.1% Saponin, secondary antibody and DAPI diluted in the same blocking buffer were incubated for one hour at room temperature. Slices were mounted with Mowiol (Millipore #475904) in superfrost slides (Thermo Fisher Scientific) and #1.5 coverslips.

Imaging and image analysis

Confocal images were acquired in a Leica TCS SP8 with 20x (0.8 NA) and 63x objectives (1.4 NA). Images were taken with a pixel size of 90 nm. Widefield imaging was performed in a Zeiss AxioVision fluorescence microscope and a 3DHISTECH panoramic MIDI II digital slide scanner. Settings were adjusted for each individual experiment and kept for all conditions. Image processing was performed using FIJI (Schindelin et al., 2012), CaseViewer and Hyugens (SVI, The Netherlands) software. For Thbs4, Nestin and DCX immunofluorescence, wholemount SVZ preparations and AAV-HAS2 analysis, tile-scan was done using the navigator plug-in in the LasX software. For Nestin immunofluorescence, images were deconvoluted using the Hyugens software.

SVZ images were reconstructed using the tile-scan tool (LasX software, Leica). SVZ region of interest (ROI) was delimited manually and expression of each marker was measured in the SVZ ROI. Quantification of all markers was done manually, except for HABP quantification, which was done with custom FIJI scripts for in vitro (https://github.com/MariaArdaya/Hyaluronan_v2) and in vivo analysis (<https://github.com/SoriaFN/Tools>). The HABP signal was segmented with an automatic threshold and colocalized with other markers such as CD44. This, combined with cell segmentation, based on the Thbs4 mask, allowed for quantification of HABP area and number of spots inside cells, outside cells or at the membrane (with a band ROI of 1 µm). The Analyze Skeleton and Fractal count plugins were used on the skeletonized segmented HABP to quantify the longest shortest path and the fractal dimension, respectively, to reveal the length and interconnectivity of the matrix network (Casey et al., 2024; Soria et al., 2020).

Western blot

The SVZ was freshly dissected on ice and minced using a micro grinder. Total protein was extracted after tissue lysis with RIPA buffer (Thermo Fisher Scientific #89900) and Protease and Phosphatase Inhibitor Cocktail 100X (Thermo Fisher Scientific). Samples were lysed in ice for 10 minutes and sonicated using an Ultrasonic Processor (Hielscher Ultrasound Technology). Sonicated tissues were centrifuged at 1200 rpm for 10 min and 4°C to remove insoluble fragments. Protein concentration was measured by chemiluminescence using RC DC™ Protein Assay (Bio-Rad). Forty micrograms of protein samples were denatured in reducing loading buffer containing β-mercaptoethanol (M3148, Sigma) and 0.0002% bromophenol blue at 95°C for 5 minutes and separated in pre-casted 12% or 7.5% Tris-Glycine polyacrylamide gel using the Criterion cell system (Biorad). Electrophoresis was conducted at 80 V in a Tris-Glycine buffer (25 mM Tris, 192 mM glycine, 0.1% SDS in dH₂O, pH 8.3). After complete separation, proteins were transferred to a nitrocellulose membrane (Biorad) using Trans-Blot® Turbo™ Transfer System (Bio-Rad). Nitrocellulose membranes were incubated in blocking solution (Tris-Buffer Solution (TBS), 0.1% Tween-20 and 5% BSA) for one hour at room temperature. Primary antibodies were incubated overnight at 4°C. Horseradish peroxidase-conjugated secondary antibodies were incubated in blocking buffer for one hour at room temperature. Immunodetection was performed by electrochemical luminescence in a Chemidoc MP (Biorad) and the intensity of the bands was quantified using ImageLab (Bio-Rad) software corrected by Gaussian curves.

Table 1.

Primary antibodies used for in vivo immunofluorescence protocol

Antibody	Host	Isotope	Dilution	Reference	Company
Thbs4	Mouse	Monoclonal IgM kappa light chain	1/400	Sc-390734	Santa Cruz Biotechnology, USA.
Thbs4	Goat	Polyclonal IgG	1/200	AF2390	R&B systems, USA
S100 β	Rabbit	Polyclonal	1/400	Z0311	Dako, Agilent Technologies, Inc.
S100 β	Mouse	Monoclonal IgG2a	1/1000	MA5-12969	Invitrogen, Spain
GFAP	Mouse	Monoclonal IgG1	1/1000	MAB3402	Sigma, Spain
GFAP	Rabbit	Polyclonal	1/4000	Z0334	Dako, Agilent Technologies, Inc.
β III tubulin	Rabbit	Polyclonal	1/400	ab18207	Abcam, UK
NeuN	Mouse	Monoclonal IgG1	1/1000	MAB377	Millipore, Germany
DCX	Rabbit	Polyclonal IgG	1/500	ab18723	Abcam, UK
Nestin	Chicken	Polyclonal IgY	1/1000	NES	Aves Lab, EEUU
Prominin 1	Rat	Monoclonal IgG1 kappa	1/300	14-1331-82	Invitrogen, Spain
EGFR	Mouse	Monoclonal	1/400	SAB4200809	Abcam, UK
BrdU	Rat	Monoclonal	1/400	MCA2060	Bio-Rad, USA.
BrdU	Mouse	Monoclonal IgG1, kappa	1/500	03-3900	Invitrogen, Spain
IdU	Mouse	Monoclonal IgG2b	1/100	MA5-24879	Invitrogen, Spain
Ki67	Rabbit	Polyclonal IgG	1/400	Ab15580	Abcam, UK
Cleaved-Caspase 3	Rabbit	Monoclonal IgG	1/300	#9664	Cell Signaling, Germany
CD44	Rat	Monoclonal IgG2b	1/500	MA517875	Invitrogen, Spain
FLAG	Mouse	Monoclonal IgG1	1/500	F1804	Sigma, Spain
Olig2	Mouse	Monoclonal IgG2ac	1/1000	MABN50	Millipore, Germany

Table 2.

Primer sequences for qPCR

Name	Sequence
Hyal1_FW	GTGCCAAGCCCTATGCTAATAAG
Hyal1_REV	GCATGTCCATTGCAAAGACTGA
Hyal2_FW	GTCCCACATACACCCGAGGA
Hyal2_REV	GGCACTCTCACCAGTGGTAGA
Hyal3_Ffw	GGACGACCTGATGCAGACTATTG
Hyal3_REV	GGTCCCCCAGAGTACCACT
Has1_FW	AGGGCTCTTAAAGGAGGAGTCC
Has1_REV	AGAAGGTAAACTGAGTCCCCAGAA
Has2_FW	CAAAGAGGTTTCGTTCAAGTTCTGA
Has2_REV	TGTGTTTGTTCCTCACTAGCTCTC
Hif1a_FW	CATAAAGTCTGCAACATGGAAGGT
Hif1a_REV	ATTTGATGGGTGAGGAATGGGTT
HPRT1_FW	TGATAGATCCATTCTATGACTGTAGA
HPRT1_REV	AAGACATTCTTTCCAGTTAAAGTTGAG

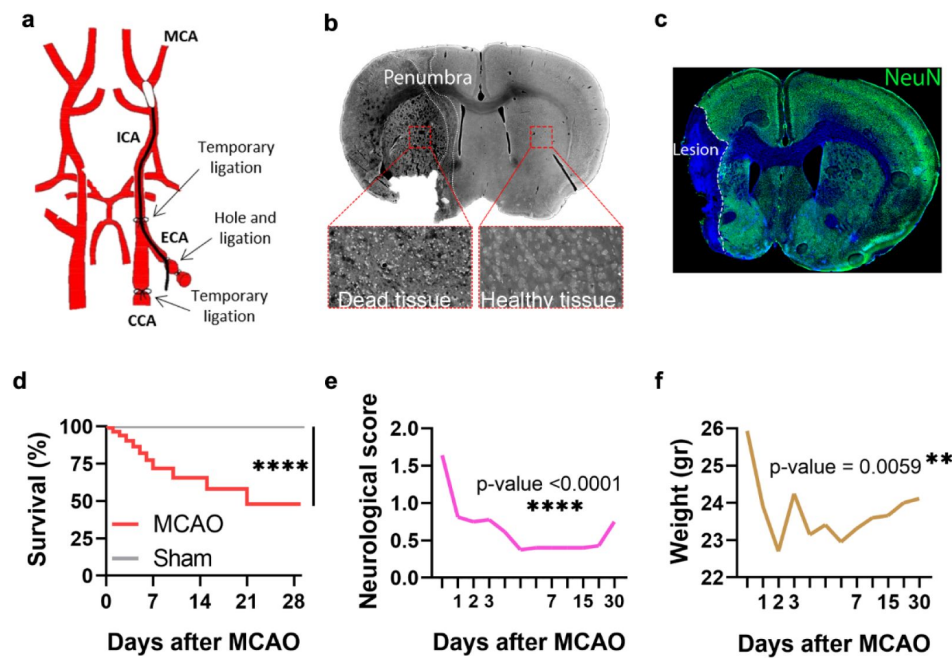
qRT-PCR

Fresh SVZ tissue from ipsilateral hemisphere was isolated as previously described from sham, 15 and 30 days after MCAO animals to analyze the expression of several genes (Supplementary Table 2). After mechanical trituration, RNA was extracted from the tissue after mechanical trituration following the manufacturer's protocol (NZYTECH Total RNA Isolation kit protocol (NZYTECH, MB13402). RNA concentration was measured by spectrophotometry using the Nanodrop 2000c spectrophotometer (Thermo-Scientific, USA). Real time polymerase chain reaction (RT-PCR) from 10 ng of RNA was performed using the One-step NZYTECH RT-qPCR Green kit according to manufacturer's protocol (NZYTECH, MB34302). RT-PCR was normalized using the housekeeping genes hypoxanthine phosphoribosyl-transferase 1 (HPRT1, Qiagen QuantiTect Primer Assay, Barcelona, Spain). Specificity of each amplification was confirmed by melting curve analysis. PCR reactions were performed in a CFX96 Detection System (BioRad, Madrid, Spain). Semi-quantification was performed using the $2^{-\Delta\Delta C_t}$ algorithm. Unsupervised hierarchical distance matrix and distance matrix map was developed using Orange3 software (Python v3.0).

Statistical analysis

Statistical analysis was done using GraphPad Prism software (version 8.0; GraphPad software). Data are presented as mean $\pm$ SEM unless specified otherwise. Statistical analysis was performed using unpaired Student's t-test for independent conditions and paired Student's t-test for dependent conditions. In addition, one-way and two-way ANOVA were also performed to compare more than 2 groups among them. Tukey and Dunnett post-hoc tests were used. For HABP skeleton analysis, Kruskal-Wallis analysis was performed to compare cumulative frequency curves in more than 2 groups. *P* value was considered significant at $p < 0.05$. Sample number (*n*), *p* values and statistical test are indicated in figure legends.

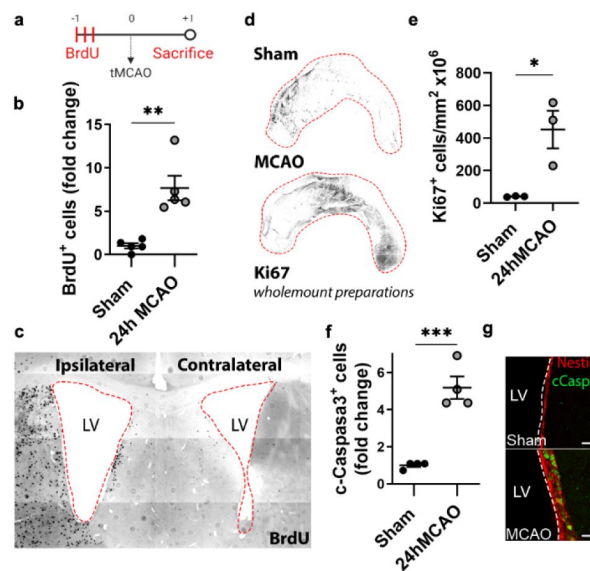
Supplementary information



Supplementary Figure 1

- Characterization of the Middle Cerebral Artery Occlusion (MCAO) mouse model of brain ischemia

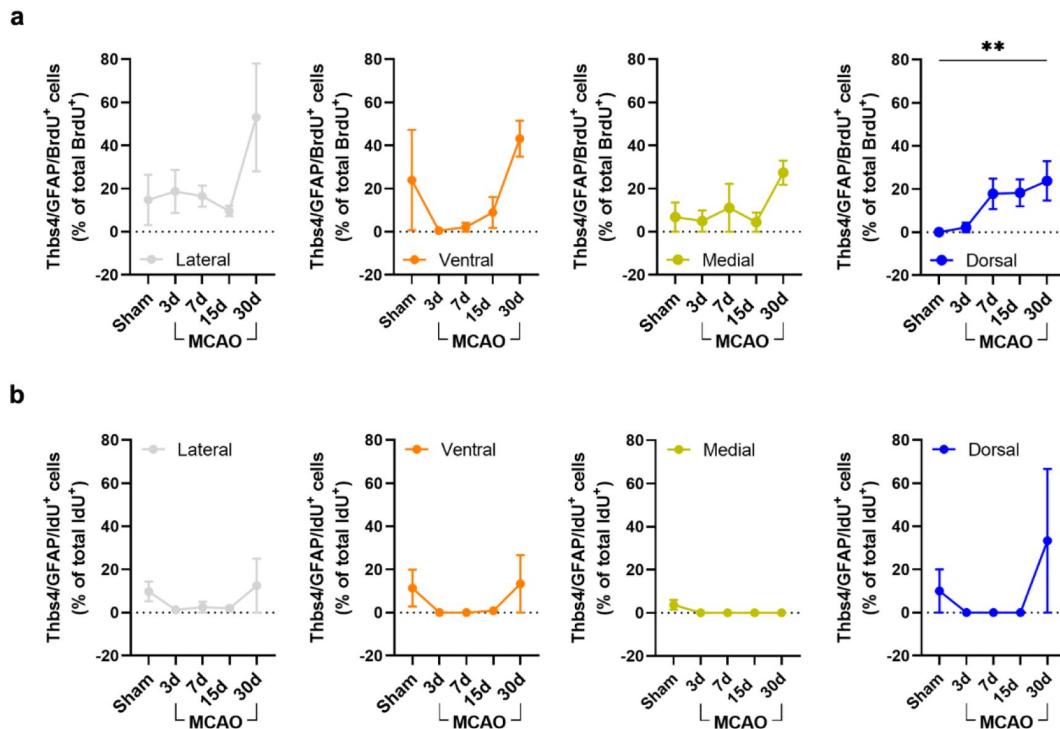
Related to [Figure 2](#). (a) Scheme of the procedure: 10 mm silicon-coated filament was inserted through an external artery and re-directed to the internal one. Ischemia was induced after 60 minutes of occlusion. After that, the filament was removed, and animals were observed every day before euthanasia. (b) Volume of infarction was observed by cresyl violet staining. (c) Neuronal death was observed by NeuN immunofluorescence. (d) Around 50% of ischemic mice survived at 28 dpi. (e) Ischemic animals ameliorated neurological symptoms over time. (f) Weight was measured as controls for mice healthcare. $n = 17$ (d, e, f). $^{**}p < 0.01$ and $^{****}p < 0.0001$ by survival curve test and Tukey post-hoc test (after one-way ANOVA was significant at $p < 0.05$).



Supplementary Figure 2

- Cell proliferation in the SVZ after brain ischemia

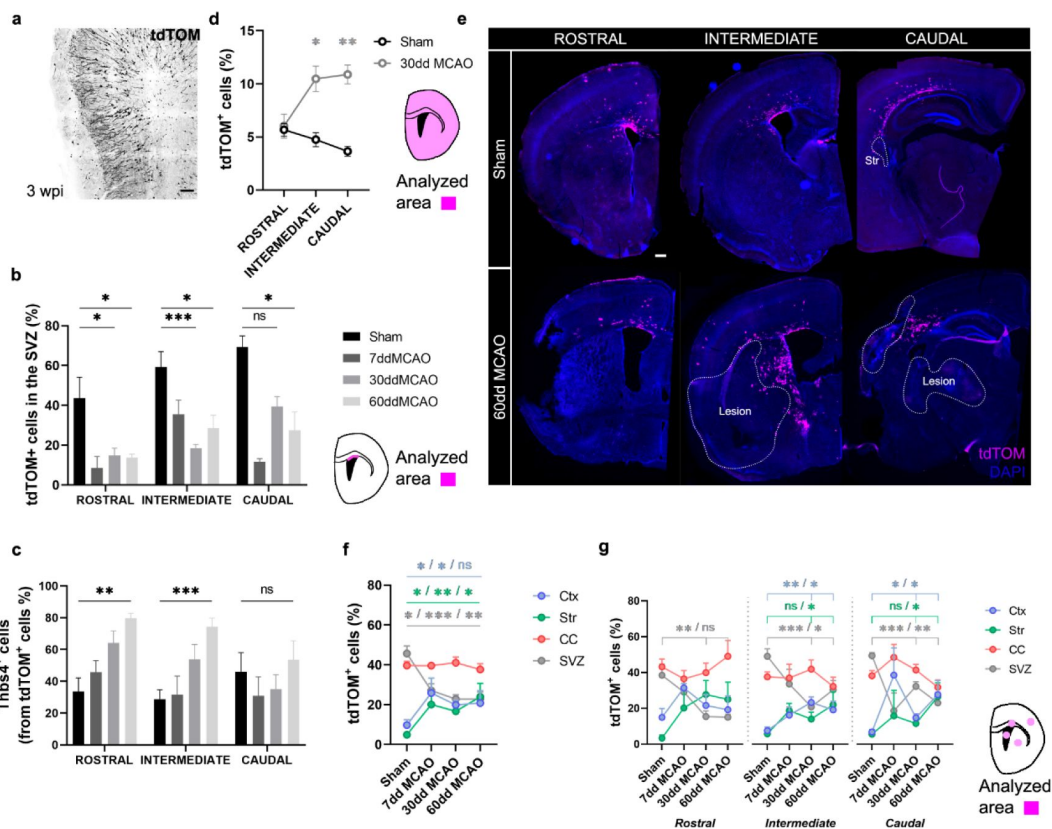
Related to [Figure 2](#). (a) Experimental design: three BrdU (50mg/kg) injections were performed the day before MCAO protocol. Animals were sacrificed 24 hours after the brain stroke model. (b) BrdU-positive cells increased in the SVZ 24 hours after MCAO. (c) Representative image of BrdU positive cells in the ipsilateral (left) and contralateral (right) hemisphere 24 hours after MCAO. (d) Ki67 marker was also used to analyze cell proliferation in wholemount preparations of the SVZ. (e) Ki67 positive cells increased in the SVZ 24 hours after MCAO. (f) The Cleaved-Caspase 3 marker increased in the SVZ 24 hours after MCAO. (g) Cleaved-caspase 3 marker was used to assess apoptosis in the SVZ after brain ischemia. Scale bar = 15 μ m (g). n = 5 (sham) and 4 (MCAO). Bars represent mean $\pm$ SEM. * p < 0.05, ** p < 0.01 and *** p < 0.001 by two-tail Student's t-test (sham vs. 24h MCAO).



Supplementary Figure 3

- Thbs4 astrocytes derive from slow proliferative cells after brain ischemia

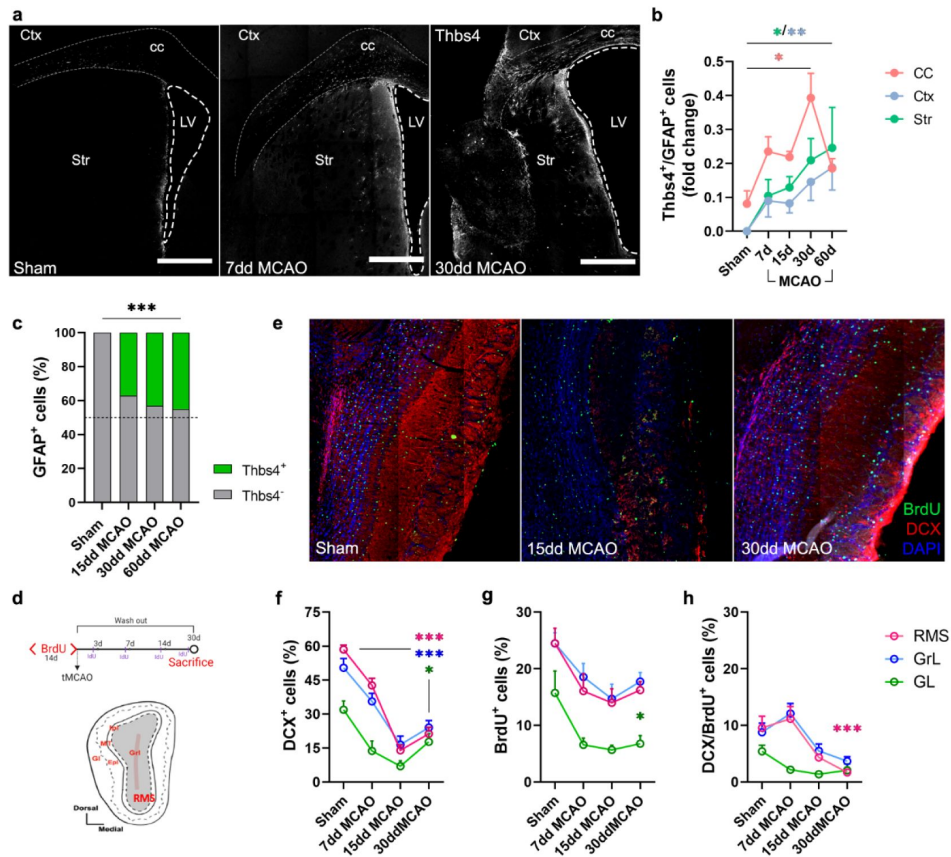
Related to **Figure 3**. (a) Slow proliferative cells were quantified in the lateral (grey), ventral (orange), medial (yellow) and dorsal (blue) SVZ. We only observed an increase of Thbs4/GFAP/BrdU positive cells in the dorsal SVZ 30 dpi. (b) Thbs4/GFAP/IdU positive cells did not show changes in the lateral (grey), ventral (orange), medial (yellow) and dorsal (blue) SVZ after brain ischemia. $n = 4$ (sham) and 3 (MCAO). $**p < 0.01$ by two-tail Student's t -test (sham vs. 30dd MCAO).



Supplementary Figure 4

- Ischemic-induced Thbs4 astrocytes migrate preferentially to caudal infarcted areas

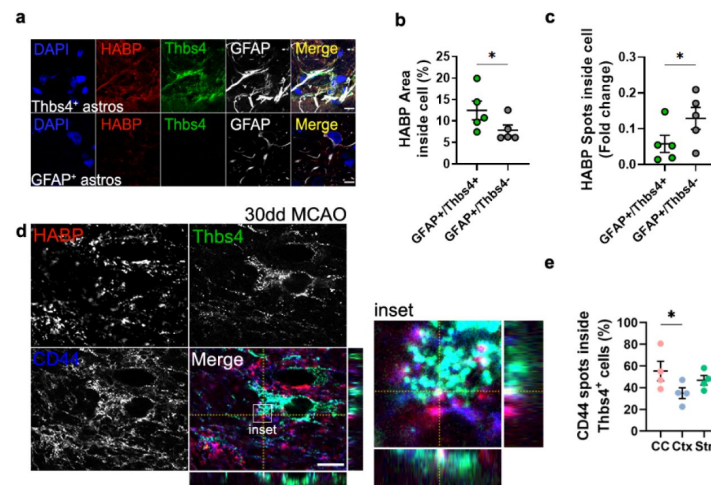
Related to [Figure 4](#). (a) Representative image of tdTOM positive cells in the OB three weeks after electroporation (3wpi). (b) tdTOM-positive cells decreased in the rostral, intermediate and caudal SVZ 7, 30 and 60 dpi. (c) Thbs4/tdTOM positive cells increased in rostral and intermediate SVZ 30 and 60 dpi. (d) Expression of tdTomato increased in intermediate and caudal brain areas at 30 dpi. (e) Representative images of tdTOM expression in rostral (left), intermediate (middle) and caudal (right) brain areas of sham (top) and 60 dpi mice (bottom). (f) tdTOM positive cells increased in the infarcted cortex (Ctx) and striatum (Str) whereas decreased in the dorsal SVZ (SVZ). No changes were observed in the corpus callosum (CC). (g) tdTOM positive cells only increased in the infarcted cortex and striatum at intermediate and caudal brain regions. Scale bar = 100 μ m (a) and 1000 μ m (e). n = 6 animals per condition. Bars represent mean $\pm$ SEM. * p < 0.05, ** p < 0.01 and *** p < 0.001 by Dunnett/Tukey post-hoc test (after two-way ANOVA was significant at p < 0.05).



Supplementary Figure 5

- SVZ NSC stop producing newborn neurons to send newborn astrocytes to the ischemic areas

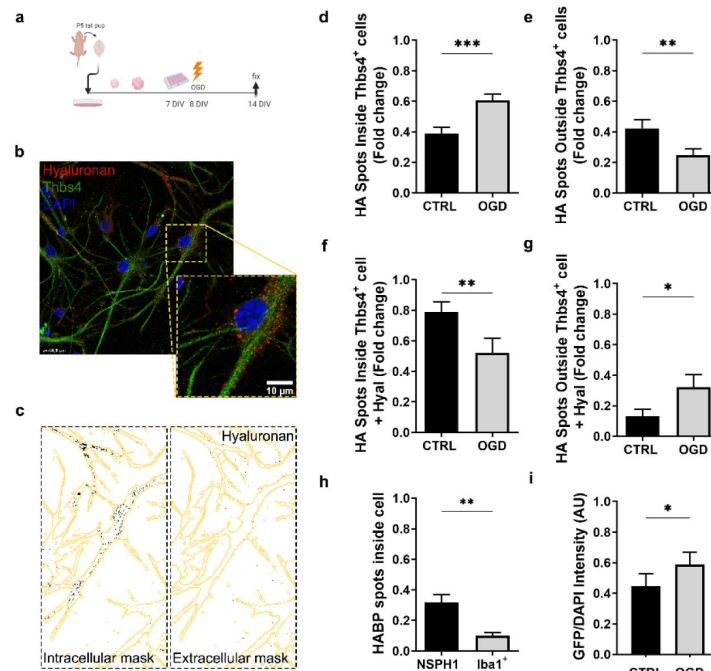
Related to **Figure 5**. (a) Thbs4 expression in the corpus callosum (cc), infarcted cortex (Ctx) and infarcted striatum (Str) of sham and 7 and 30 dpi mice. (b) Thbs4 increased by 30 dpi in the corpus callosum (cc) and 60 dpi in the infarcted cortex (Ctx) and infarcted striatum (Str). (c) Expression of Thbs4 increased in total GFAP population after 15, 30 and 60 dpi. (d) Experimental design: chronic BrdU (1% in water) treatment was administered for two weeks. BrdU only persisted in slow proliferative NSC (Codega et al., 2014). DCX and BrdU positive cells were measured in the OB layers 7, 15 and 30 dpi. (e) Representative images of BrdU and DCX marker in sham (left), 15 (middle) and 30 (right) dpi mice OB. (f) DCX positive cells decreased 15 and 30 dpi in the RMS, granular OB layer (GrL) and glomerular OB layer (GL). (g) BrdU positive cells only decreased 30 dpi in the glomerular OB layer (GL). Double DCX/BrdU positive cells significantly decreased in the RMS 30 dpi. In (f) and (g), values are normalized to total DAPI nuclei (100%). Scale bar = 500 μ m (a). $n = 6$ animals per condition. Lines represent mean $\pm$ SEM. $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$ by Chi-square test and Tukey post-hoc test (after two-way ANOVA was significant at $p < 0.05$).



Supplementary Figure 6

– Thbs4 astrocytes internalize HA in the infarcted areas

Related to **Figure 6** [\[link\]](#). (a) Representative images of Thbs4/GFAP (“newborn astrocytes”, top) and GFAP-only (“local astrocytes”, bottom) cells in the infarcted areas at 30 dpi. (b) Thbs4-positive astrocytes internalized more HA compared to local astrocytes only labelled with GFAP marker. (c) However, the number of HA spots internalized was higher in local astrocytes compared with Thbs4 astrocytes, suggesting that Thbs4 internalize HA in fewer vesicles. (d) Representative images of HABP (red), Thbs4 (green) and CD44 (blue) at the lesion border. Inset shows an orthogonal projection of Thbs4/CD44/HABP-positive intracellular vesicle. (e) Thbs4 positive astrocytes expressed a high rate of CD44 receptors so much in corpus callosum (CC), infarcted cortex (Ctx) as in infarcted striatum (Str). Scale bar = 10 μ m (c, f). n = 5 animals per condition. Bars represent mean $\pm$ SEM. * p < 0.05 and ** p < 0.01 by Tukey posthoc test (after two-way ANOVA was significant at p < 0.05) and two-tail Student’s t-test (Thbs4 vs GFAP).



Supplementary Figure 7

- In vitro NSC-derived Thbs4 cells synthesize HA after a hypoxic-ischemic condition

Related to [Figure 6](#). (a) Experimental design: P4 pup rats were sacrificed, and SVZ NSC were extracted and cultured in the proliferative medium for one week. After that, neurospheres were disaggregated and seeded isolated in a culture dish. The following day, oxygen and glucose deprivation protocol (OGD) was performed. Cells were fixed 7 days after culture them. (b) Representative images of in vitro Thbs4 positive cells (green) and HA (red). (c) HA spots were measured inside and outside Thbs4 positive cells. (d) HA spots increased inside Thbs4 positive astrocytes after OGD whereas (e) decreased in the extracellular space. However, when NSC were exposed to hyaluronidases in order to remove extracellular HA, Thbs4 positive cells decreased internalization of HA spots (f) and increased extracellular HA spots (g). (h) Neurosphere-derived NSC (NSPH1) internalized more HA spots compared with Iba1-positive cells. (i) Metalloproteinase activity was higher in OGD NSC cultures compared to control condition. Scale bar = 10 µm (b). n = 15 samples per condition. Bars represent mean ± SEM. **p* < 0.05, ***p* < 0.01 and ****p* < 0.001 by two-tail Student's t-test (CTRL vs. OGD; NSPH1 vs. Iba1⁺).

Acknowledgements

The authors thank Juan Carlos Chara, Zara Martínez-Paez, and Laura Escobar for their invaluable technical support. We thank Laura Simon for proofreading the manuscript, and Abraham Martín, Mónica Benito-Muñoz and Sol Beccari for training in MCAO in mice. We also thank Vera Gorbunova (University of Rochester, USA) for providing the cDNA of HAS2 from the naked mole rat. We thank Marie-Laure Thiolat and Dr. Nathalie Dutheil for AAV production. This study received financial support from the French government in the framework of the University of Bordeaux's IdEx "Investments for the Future" program/GPR BRAIN_2030 (B.D.). F.C. acknowledges funding from BIOEF (BIO21/COV/012), Eukampus, Plan Complementario en Salud (PRTR-C17.I1) and the Basque Government. F.N.S. was funded by the Spanish Research Agency (IJCI-2017-32114, PID2020-115896RJ-I00 and RYC2021-032602-I co-funded by NextGeneration EU/PRTR). C.M. acknowledges funding from the Basque Government (IT1203-19) and CIBERNED. M.A. was supported by a PhD fellowship from the Basque Government.

Additional information

Author contributions

M.A. conducted experiments and collected and analyzed data. M.C.T. and H.C. provided the Ai14-Rosa26-CAG-tdTomato^{flx/flx} mice and technical input for electroporation. B.D. provided the AAV-HAS2 and conceived the idea of glial scar simulation. F.C., F.N.S. and C.M. secured funding. C.M. provided infrastructural support and scientific feedback. F.C. and F.N.S. conceived and supervised the study. F.P.C. supervised the study. M.A. and F.N.S. prepared the figures. M.A., F.N.S. and F.C. wrote the paper, with input from all authors.

References

- Adams KL, Gallo V (2018) **The diversity and disparity of the glial scar** *Nat Neurosci* **21**:9–15 <https://doi.org/10.1038/s41593-017-0033-9>
- Anderson MA, Burda JE, Ren Y, Ao Y, O’Shea TM, Kawaguchi R, Coppola G, Khakh BS, Deming TJ, Sofroniew MV (2016) **Astrocyte scar formation aids central nervous system axon regeneration** *Nature* **532**:195–200 <https://doi.org/10.1038/nature17623>
- Arvidsson A, Collin T, Kirik D, Kokaia Z, Lindvall O (2002) **Neuronal replacement from endogenous precursors in the adult brain after stroke** *Nat Med* **8**:963–970 <https://doi.org/10.1038/nm747>
- Basak O *et al.* (2018) **Troy+ brain stem cells cycle through quiescence and regulate their number by sensing niche occupancy** *Proc Natl Acad Sci U S A* **115**:E610–E619 <https://doi.org/10.1073/pnas.1715911114>
- Beckervordersandforth R *et al.* (2010) **In vivo fate mapping and expression analysis reveals molecular hallmarks of prospectively isolated adult neural stem cells** *Cell Stem Cell* **7**:744–758 <https://doi.org/10.1016/j.stem.2010.11.017>
- Benito-Muñoz M, Matute C, Cavaliere F (2016) **Adenosine A1 receptor inhibits postnatal neurogenesis and sustains astroglialogenesis from the subventricular zone** *Glia* **64**:1465–1478 <https://doi.org/10.1002/glia.23010>
- Benner EJ, Luciano D, Jo R, Abdi K, Paez-Gonzalez P, Sheng H, Warner DS, Liu C, Eroglu C, Kuo CT (2013) **Protective astrogenesis from the SVZ niche after injury is controlled by Notch modulator Thbs4** *Nature* **497**:369–373 <https://doi.org/10.1038/nature12069>
- Bond AM, Ming G-L, Song H (2015) **Adult Mammalian Neural Stem Cells and Neurogenesis: Five Decades Later** *Cell Stem Cell* **17**:385–395 <https://doi.org/10.1016/j.stem.2015.09.003>
- Bonzano S, Crisci I, Podlesny-Drabiniok A, Rolando C, Krezel W, Studer M, De Marchis S. (2018) **Neuron-Astroglia Cell Fate Decision in the Adult Mouse Hippocampal Neurogenic Niche Is Cell-Intrinsically Controlled by COUP-TFI In Vivo** *Cell Rep* **24**:329–341 <https://doi.org/10.1016/j.celrep.2018.06.044>
- Bourdenx M *et al.* (2015) **Lack of additive role of ageing in nigrostriatal neurodegeneration triggered by α -synuclein overexpression** *Acta Neuropathol Commun* **3** <https://doi.org/10.1186/s40478-015-0222-2>
- Bush TG, Puvanachandra N, Horner CH, Polito A, Ostenfeld T, Svendsen CN, Mucke L, Johnson MH, Sofroniew MV (1999) **Leukocyte infiltration, neuronal degeneration, and neurite outgrowth after ablation of scar-forming, reactive astrocytes in adult transgenic mice** *Neuron* **23**:297–308 [https://doi.org/10.1016/s0896-6273\(00\)80781-3](https://doi.org/10.1016/s0896-6273(00)80781-3)
- Cai H, Ma Y, Jiang L, Mu Z, Jiang Z, Chen X, Wang Y, Yang G-Y, Zhang Z (2017) **Hypoxia Response Element-Regulated MMP-9 Promotes Neurological Recovery via Glial Scar Degradation and Angiogenesis in Delayed Stroke** *Molecular Therapy* **25**:1448–1459 <https://doi.org/10.1016/j.ymthe.2017.03.020>

- Casey DT, Lahue KG, Mori V, Herrmann J, Hall JK, Suki B, Janssen-Heininger YMW, Bates JHT (2024) **Local fractal dimension of collagen detects increased spatial complexity in fibrosis** *Histochem Cell Biol* **161**:29–42 <https://doi.org/10.1007/s00418-023-02248-8>
- Cavaliere F, Dinkel K, Reymann K (2006) **The subventricular zone releases factors which can be protective in oxygen/glucose deprivation-induced cortical damage: an organotypic study** *Exp Neurol* **201**:66–74 <https://doi.org/10.1016/j.expneurol.2006.03.020>
- Ceanga M, Dahab M, Witte OW, Keiner S (2021) **Adult Neurogenesis and Stroke: A Tale of Two Neurogenic Niches** *Frontiers in Neuroscience* **15**
- Cebrian-Silla A *et al.* (2021) **Single-cell analysis of the ventricular-subventricular zone reveals signatures of dorsal and ventral adult neurogenesis** *eLife* **10** <https://doi.org/10.7554/eLife.67436>
- Chaker Z, Codega P, Doetsch F (2016) **A mosaic world: puzzles revealed by adult neural stem cell heterogeneity** *WIREs Developmental Biology* **5**:640–658 <https://doi.org/10.1002/wdev.248>
- Codega P, Silva-Vargas V, Paul A, Maldonado-Soto AR, DeLeo AM, Pastrana E, Doetsch F (2014) **Prospective Identification and Purification of Quiescent Adult Neural Stem Cells from Their In Vivo Niche** *Neuron* **82**:545–559 <https://doi.org/10.1016/j.neuron.2014.02.039>
- Cregg JM, DePaul MA, Filous AR, Lang BT, Tran A, Silver J (2014) **Functional regeneration beyond the glial scar** *Exp Neurol* **253**:197–207 <https://doi.org/10.1016/j.expneurol.2013.12.024>
- Doetsch F, Caillé I, Lim DA, García-Verdugo JM, Alvarez-Buylla A (1999) **Subventricular zone astrocytes are neural stem cells in the adult mammalian brain** *Cell* **97**:703–716 [https://doi.org/10.1016/s0092-8674\(00\)80783-7](https://doi.org/10.1016/s0092-8674(00)80783-7)
- Dzwonek J, Wilczynski GM (2015) **CD44: molecular interactions, signaling and functions in the nervous system** *Front Cell Neurosci* **9** <https://doi.org/10.3389/fncel.2015.00175>
- Dzyubenko E, Manrique-Castano D, Kleinschnitz C, Faissner A, Hermann DM (2018) **Role of immune responses for extracellular matrix remodeling in the ischemic brain** *Ther Adv Neurol Disord* **11** <https://doi.org/10.1177/1756286418818092>
- Encinas JM, Michurina TV, Peunova N, Park J-H, Tordo J, Peterson DA, Fishell G, Koulakov A, Enikolopov G (2011) **Division-coupled astrocytic differentiation and age-related depletion of neural stem cells in the adult hippocampus** *Cell Stem Cell* **8**:566–579 <https://doi.org/10.1016/j.stem.2011.03.010>
- Faulkner JR, Herrmann JE, Woo MJ, Tansey KE, Doan NB, Sofroniew MV (2004) **Reactive astrocytes protect tissue and preserve function after spinal cord injury** *J Neurosci* **24**:2143–2155 <https://doi.org/10.1523/JNEUROSCI.3547-03.2004>
- Fawcett JW (2015) **The extracellular matrix in plasticity and regeneration after CNS injury and neurodegenerative disease** *Prog Brain Res* **218**:213–226 <https://doi.org/10.1016/bs.pbr.2015.02.001>
- Fawcett JW, Asher RA (1999) **The glial scar and central nervous system repair** *Brain Res Bull* **49**:377–391 [https://doi.org/10.1016/s0361-9230\(99\)00072-6](https://doi.org/10.1016/s0361-9230(99)00072-6)

- Frisén J, Johansson CB, Török C, Risling M, Lendahl U (1995) **Rapid, widespread, and longlasting induction of nestin contributes to the generation of glial scar tissue after CNS injury** *Journal of Cell Biology* **131**:453–464 <https://doi.org/10.1083/jcb.131.2.453>
- Gelderblom M *et al.* (2009) **Temporal and spatial dynamics of cerebral immune cell accumulation in stroke** *Stroke* **40**:1849–1857 <https://doi.org/10.1161/STROKEAHA.108.534503>
- Girard F, Eichenberger S, Celio MR (2014) **Thrombospondin 4 deficiency in mouse impairs neuronal migration in the early postnatal and adult brain** *Molecular and Cellular Neuroscience* **61**:176–186 <https://doi.org/10.1016/j.mcn.2014.06.010>
- Jayaraj RL, Azimullah S, Beiram R, Jalal FY, Rosenberg GA (2019) **Neuroinflammation: friend and foe for ischemic stroke** *J Neuroinflammation* **16** <https://doi.org/10.1186/s12974-019-1516-2>
- Kjell J *et al.* (2020) **Defining the Adult Neural Stem Cell Niche Proteome Identifies Key Regulators of Adult Neurogenesis** *Cell Stem Cell* **26**:277–293
- Kralic JE, Ledergerber DA, Fritschy J-M (2005) **Disruption of the neurogenic potential of the dentate gyrus in a mouse model of temporal lobe epilepsy with focal seizures** *European Journal of Neuroscience* **22**:1916–1927 <https://doi.org/10.1111/j.1460-9568.2005.04386.x>
- Lau LW, Cua R, Keough MB, Haylock-Jacobs S, Yong VW (2013) **Pathophysiology of the brain extracellular matrix: a new target for remyelination** *Nat Rev Neurosci* **14**:722–729
- Laug D *et al.* (2019) **Nuclear factor I-A regulates diverse reactive astrocyte responses after CNS injury** *J Clin Invest* **129**:4408–4418 <https://doi.org/10.1172/JCI127492>
- Li L, Harms KM, Ventura PB, Lagace DC, Eisch AJ, Cunningham LA (2010) **Focal cerebral ischemia induces a multilineage cytogenic response from adult subventricular zone that is predominantly gliogenic** *Glia* **58**:1610–1619 <https://doi.org/10.1002/glia.21033>
- Lindwall C, Olsson M, Osman AM, Kuhn HG, Curtis MA (2013) **Selective expression of hyaluronan and receptor for hyaluronan mediated motility (Rhamm) in the adult mouse subventricular zone and rostral migratory stream and in ischemic cortex** *Brain Res* **1503**:62–77 <https://doi.org/10.1016/j.brainres.2013.01.045>
- Liu XS, Zhang ZG, Zhang RL, Gregg S, Morris DC, Wang Y, Chopp M (2007) **Stroke induces gene profile changes associated with neurogenesis and angiogenesis in adult subventricular zone progenitor cells** *J Cereb Blood Flow Metab* **27**:564–574 <https://doi.org/10.1038/sj.jcbfm.9600371>
- Llorens-Bobadilla E, Zhao S, Baser A, Saiz-Castro G, Zwadlo K, Martin-Villalba A (2015) **Single-Cell Transcriptomics Reveals a Population of Dormant Neural Stem Cells that Become Activated upon Brain Injury** *Cell Stem Cell* **17**:329–340 <https://doi.org/10.1016/j.stem.2015.07.002>
- Long KR, Huttner WB (2019) **How the extracellular matrix shapes neural development** *Open Biol* **9**
- Luskin MB (1993) **Restricted proliferation and migration of postnatally generated neurons derived from the forebrain subventricular zone** *Neuron* **11**:173–189 [https://doi.org/10.1016/0896-6273\(93\)90281-u](https://doi.org/10.1016/0896-6273(93)90281-u)

- Muppala S, Xiao R, Krukovets I, Verbovetsky D, Yendamuri R, Habib N, Raman P, Plow E, Stenina-Adognravi O (2017) **Thrombospondin-4 mediates TGF- β -induced angiogenesis** *Oncogene* **36**:5189–5198 <https://doi.org/10.1038/onc.2017.140>
- Nguyen PT *et al.* (2020) **Microglial Remodeling of the Extracellular Matrix Promotes Synapse Plasticity** *Cell* **182**:388–403
- Platel J-C *et al.* (2019) **Neuronal integration in the adult mouse olfactory bulb is a non-selective addition process** *eLife* **8** <https://doi.org/10.7554/eLife.44830>
- Pous L *et al.* (2020) **Fibrinogen induces neural stem cell differentiation into astrocytes in the subventricular zone via BMP signaling** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-14466-y>
- Preston M, Sherman LS (2011) **Neural Stem Cell Niches: Critical Roles for the Hyaluronan-Based Extracellular Matrix in Neural Stem Cell Proliferation and Differentiation** *Front Biosci (Schol Ed)* **3**:1165–1179
- Schindelin J *et al.* (2012) **Fiji: an open-source platform for biological-image analysis** *Nat Methods* **9**:676–682
- Segel M *et al.* (2019) **Niche stiffness underlies the ageing of central nervous system progenitor cells** *Nature* **573**:130–134
- Sherman LS, Matsumoto S, Su W, Srivastava T, Back SA (2015) **Hyaluronan Synthesis, Catabolism, and Signaling in Neurodegenerative Diseases** *International Journal of Cell Biology* **2015**
- Silver J, Miller JH (2004) **Regeneration beyond the glial scar** *Nat Rev Neurosci* **5**:146–156 <https://doi.org/10.1038/nrn1326>
- Sohn J, Orosco L, Guo F, Chung S-H, Bannerman P, Mills Ko E, Zarbalis K, Deng W, Pleasure D (2015) **The subventricular zone continues to generate corpus callosum and rostral migratory stream astroglia in normal adult mice** *J Neurosci* **35**:3756–3763 <https://doi.org/10.1523/JNEUROSCI.3454-14.2015>
- Soria FN *et al.* (2020) **Synucleinopathy alters nanoscale organization and diffusion in the brain extracellular space through hyaluronan remodeling** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-17328-9>
- Sundholm-Peters NL, Yang HKC, Goings GE, Walker AS, Szele FG (2005) **Subventricular Zone Neuroblasts Emigrate Toward Cortical Lesions** *Journal of Neuropathology & Experimental Neurology* **64**:1089–1100 <https://doi.org/10.1097/01.jnen.0000190066.13312.8f>
- Tian X, Azpurua J, Hine C, Vaidya A, Myakishev-Rempel M, Ablueva J, Mao Z, Nevo E, Gorbunova V, Seluanov A (2013) **High-molecular-mass hyaluronan mediates the cancer resistance of the naked mole rat** *Nature* **499**:346–349
- Wanner IB, Anderson MA, Song B, Levine J, Fernandez A, Gray-Thompson Z, Ao Y, Sofroniew MV (2013) **Glial scar borders are formed by newly proliferated, elongated astrocytes that interact to corral inflammatory and fibrotic cells via STAT3-dependent mechanisms after spinal cord injury** *J Neurosci* **33**:12870–12886 <https://doi.org/10.1523/JNEUROSCI.2121-13.2013>

Yamashita T *et al.* (2006) **Subventricular zone-derived neuroblasts migrate and differentiate into mature neurons in the post-stroke adult striatum** *J Neurosci* **26**:6627–6636 <https://doi.org/10.1523/JNEUROSCI.0149-06.2006>

Zhang Z *et al.* (2023) **Increased hyaluronan by naked mole-rat Has2 improves healthspan in mice** *Nature* <https://doi.org/10.1038/s41586-023-06463-0>

Zimmermann DR, Dours-Zimmermann MT (2008) **Extracellular matrix of the central nervous system: from neglect to challenge** *Histochem Cell Biol* **130**:635–653 <https://doi.org/10.1007/s00418-008-0485-9>

Author information

Maria Ardaya[†]

Achucarro Basque Center for Neuroscience, Leioa, Spain, Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain, CIBERNED, Madrid, Spain
ORCID iD: [0000-0002-0103-5649](https://orcid.org/0000-0002-0103-5649)

[†]Present address: Donostia International Physics Center (DIPC), San Sebastian, Spain.

Marie-Catherine Tiveron

Aix Marseille Univ, CNRS, IBDM, Marseille, France
ORCID iD: [0000-0003-3719-3494](https://orcid.org/0000-0003-3719-3494)

Harold Cremer

Aix Marseille Univ, CNRS, IBDM, Marseille, France
ORCID iD: [0000-0002-8673-5176](https://orcid.org/0000-0002-8673-5176)

Benjamin Dehay

Univ. Bordeaux, CNRS, IMN, UMR 5293, Bordeaux, France
ORCID iD: [0000-0003-1723-9045](https://orcid.org/0000-0003-1723-9045)

Fernando Pérez-Cerdá

Achucarro Basque Center for Neuroscience, Leioa, Spain, Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain, CIBERNED, Madrid, Spain

Carlos Matute

Achucarro Basque Center for Neuroscience, Leioa, Spain, Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain, CIBERNED, Madrid, Spain
ORCID iD: [0000-0001-8672-711X](https://orcid.org/0000-0001-8672-711X)

Federico N Soria

Achucarro Basque Center for Neuroscience, Leioa, Spain, Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain, CIBERNED, Madrid, Spain, Ikerbasque, Basque Foundation for Science, Bilbao, Spain
ORCID iD: [0000-0003-1229-9663](https://orcid.org/0000-0003-1229-9663)

For correspondence: federico.soria@achucarro.org

Fabio Cavaliere*

Achucarro Basque Center for Neuroscience, Leioa, Spain, Department of Neuroscience, University of the Basque Country (UPV/EHU), Leioa, Spain, CIBERNED, Madrid, Spain, Basque Biomodel Platform for Human Research (BBioH), Leioa, Spain
ORCID iD: [0000-0003-2003-522X](https://orcid.org/0000-0003-2003-522X)

For correspondence: fabio.cavaliere@ehu.eus

*Present address: Instituto Biofisika (CSIC, UPV/EHU), Fundación Biofísica Bizkaia, Leioa, Spain

Editors

Reviewing Editor

Annalisa Scimemi

University at Albany, State University of New York, Albany, United States of America

Senior Editor

Sofia Araújo

University of Barcelona, Barcelona, Spain

Reviewer #1 (Public review):

Summary:

The authors show that SVZ derived astrocytes respond to a middle carotid artery occlusion (MCAO) hypoxia lesion by secreting and modulating hyaluronan at the edge of the lesion (penumbra) and that hyaluronin is a chemoattractant to SVZ astrocytes. They use lineage tracing of SVZ cells to determine their origin. They also find that SVZ derived astrocytes express Thbs-4 but astrocytes at the MCAO-induced scar do not. Also, they demonstrate that decreased HA in the SVZ is correlated with gliogenesis. While much of the paper is descriptive/correlative they do overexpress Hyaluronan synthase 2 via viral vectors and show this is sufficient to recruit astrocytes to the injury. Interestingly, astrocytes preferred to migrate to the MCAO than to the region of overexpressed HAS2.

Strengths:

The field has largely ignored the gliogenic response of the SVZ, especially with regards to astrocytic function. These cells and especially newborn cells may provide support for regeneration. Emigrated cells from the SVZ have been shown to be neuroprotective via creating pro-survival environments, but their expression and deposition of beneficial extracellular matrix molecules is poorly understood. Therefore, this study is timely and important. The paper is very well written and flow of result logical.

Comments on revised version:

Thanks for addressing my final points.

<https://doi.org/10.7554/eLife.96076.3.sa2>

Reviewer #2 (Public review):

Summary:

In their manuscript, Ardaya et al address the impact of ischemia-induced astrogliogenesis from the adult SVZ and their effect on remodeling of the extracellular matrix (ECM) in the glial scar. The authors show that the levels of Thbs4, a marker previously identified to be expressed in astrocytes and neural stem cells (NSCs) of the SVZ, strongly increase upon ischemia. While proliferation is significantly increase shortly after ischemia, Nestin and DCX (markers for NSCs and neuroblasts, respectively) decrease and Thbs4 levels suggesting that the neurogenic program is halted and astrogenesis is enhanced. By fate-mapping, the authors show that astrocytes derive from SVZ NSCs and migrate towards the lesion. These SVZ-derived astrocytes strongly express Thbs4 and populate the border of the lesion, while local astrocytes do not express Thbs4 and localize to both scar and border. Interestingly, the Thbs4-positive astrocytes appear to represent a second wave of astrocytes accumulating at the scar, following an immediate reaction of first wave reactive gliosis by local astrocytes. Mechanistically, the study presents evidence that the degradation of hyaluronan (HA), a key component of the extracellular matrix (ECM) is downregulated in the SVZ after ischemia, potentially inducing astrogliogenesis, while HA accumulation at the lesion side represents at least one signal to recruit the newly generated astrocytes. In the aim to facilitate tissue regeneration after ischemic injury, the authors propose that the Thbs4-positive astrocytes could be a promising therapeutical target to modulate the glial scar after brain ischemia.

Strengths:

This topic is timely and important since the focus of previous studies was almost exclusively on the role of neurogenesis. The generation of adult-born astrocytes has been proven in both neurogenic niches under physiological conditions, but the implicated function in pathology has not been sufficiently addressed yet.

Weaknesses:

The study presented by Ardaya et al presents good evidence that a population of astrocytes that express Thbs4 contribute to scar formation after ischemic injury. The authors demonstrate that ischemic injury increases proliferation in the SVZ, decreases neurogenesis and increases astrogenesis. However, whether astrogenesis is a result of terminal differentiation of type B cells or their proliferation remains unclear. Here, a combination of fate mapping and thymidine analogue-tracing would have been conclusively.

<https://doi.org/10.7554/eLife.96076.3.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Public review):

Summary:

The authors show that SVZ derived astrocytes respond to a middle carotid artery occlusion

(MCAO) hypoxia lesion by secreting and modulating hyaluronan at the edge of the lesion (penumbra) and that hyaluronan is a chemoattractant to SVZ astrocytes. They use lineage tracing of SVZ cells to determine their origin. They also find that SVZ derived astrocytes express Thbs-4 but astrocytes at the MCAO-induced scar do not. Also, they demonstrate that decreased HA in the SVZ is correlated with gliogenesis. While much of the paper is descriptive/correlative they do overexpress Hyaluronan synthase 2 via viral

vectors and show this is sufficient to recruit astrocytes to the injury. Interestingly, astrocytes preferred to migrate to the MCAO than to the region of overexpressed HAS2.

Strengths:

The field has largely ignored the gliogenic response of the SVZ, especially with regards to astrocytic function. These cells and especially newborn cells may provide support for regeneration. Emigrated cells from the SVZ have been shown to be neuroprotective via creating pro-survival environments, but their expression and deposition of beneficial extracellular matrix molecules is poorly understood. Therefore, this study is timely and important. The paper is very well written and the flow of results logical.

Comments on revised version:

The authors have addressed my points and the paper is much improved. Here are the salient remaining issues that I suggest be addressed.

We appreciate the feedback by the reviewer, and we are glad that the paper is considered to be much improved. We have done our best to address the remaining issues in this 2nd revision.

The authors have still not shown, using loss of function studies, that Hyaluronan is necessary for SVZ astrogenesis and or migration to MCAO lesions.

This is true. Unfortunately, complete removal of hyaluronan (via Hyase) triggers epilepsy, already described in 1963 by James Young (*Exp Neurol* paper). Degradation by Hyase also provokes neuroinflammation (Soria et al., 2020 Nat Commun). Two alternatives could be 1) partial depletion with Has inhibitor 4MU (but it is also associated with increased inflammation) or 2) a Has-KO mouse, such as Has3^{-/-} (Arranz et al., 2014), although, to our knowledge, this mouse line is not openly available. We have added a sentence in line 332 addressing this shortcoming: "Loss-of-function studies, using HA-depletion models or HA synthase (Has)deficient mice are still needed to corroborate this finding, although the inflammation associated with HA deficiency might confound interpretation."

(1) The co-expression of EGFr with Thbs4 and the literature examination is useful.

We thank the reviewer for the kind comment.

(2) Too bad they cannot explain the lack of effect of the MCAO on type C cells. The comparison with kainate-induced epilepsy in the hippocampus may or may not be relevant.

As stated in the previous response, we also found this interesting, and it does warrant further exploration by looking into possible direct NSC-astrocyte differentiation. But we believe that both this possible direct differentiation and the reactive status for these astrocytes are out of the scope of the study. We will not speculate about this in the discussion, either.

(3) Thanks for including the orthogonal confocal views in Fig S6D.

(4) The statement that "BrdU+/Thbs4+ cells mostly in the dorsal area" and therefore they mostly focused on that region is strange. Figure 8 clearly shows Thbs4 staining all along the striatal SVZ. Do they mean the dorsal segment of the striatal SVZ or the subcallosal SVZ? Fig. 4b and Fig 4f clearly show the "subcallosal" area as the one analysed but other figures show the dorsal striatal region (Fig. 2a). This is important because of the well-known embryological and neurogenic differences between the regions.

While it is true that Thbs4 is also expressed in the other subregions of the SVZ (lateral, ventral and medial), as observed in Fig 8. we chose the dorsal area because it is the subregion where we observed the larger increase in slow proliferative NSCs (Thbs4/GFAP/BrdU-positive cells) after MCAO (Fig S3). As observed in the quantifications in Fig S3, we found Thbs4/GFAP/BrdU-positive cells increase in lateral, medial and ventral SVZ, but it is not significant. Therefore, from Fig 4 onwards, we focused on the dorsal SVZ, which the reviewer mentions as “subcallosal” area. We chose the term “dorsal” as stated in single-cell studies (Cebrian-Silla et al, 2021, eLife; Marcy et al., 2023, Sci Adv) and reviews (Sequerra 2014 Front Cell Neurosci) that investigate or mention this subregion, respectively. In the abstract, we are perfectly clear stating that newborn astrocytes migrate from both dorsal and medial areas.

In Fig 2a, the immunofluorescence image shows medial and lateral SVZ, but at this point in the paper, we have not yet made specific subregional quantifications, and the Nestin, DCX and Thbs4 quantifications refer to the SVZ as a whole, both in the IF and in the WB (Fig 2e-g). We apologize for the confusion. We have clarified this in the text (line 119).

(5) It is good to know that the harsh MCAO's had already been excluded.

(6) Sorry for the lack of clarity - in addition to Thbs4, I was referring to mouse versus rat Hyaluronan degradation genes (Hyal1, Hyal2 and Hyal3) and hyaluronan synthase genes (HAS1 and HAS2) in order to address the overall species differences in hyaluronan biology thus justifying the "shift" from mouse to rat. You examine these in the (weirdly positioned) Fig. 8h,i. Please add a few sentences on mouse vs rat Thbs4 and Hyaluronan relevant genes.

We thank the reviewer for these remarks. We have now added a sentence pointing to the similar internalization and degradation in rat and mouse (reviewed by Sherman et al., 2015). This correction is in line 233. Hyaluronan is, in evolutionary terms, a very “old” molecule, part of the “ancient” glycan-based matrix, before the evolution of proteoglycans and fibrous proteins such as collagen, laminin etc. Hence, its machinery is highly conserved across species.

We have also reorganized the panels in Fig 8, where 8h and 8i were indeed weirdly positioned. We hope that the new version of this figure is more easily readable.

(7) Thank you for the better justification of using the naked mole rat HA synthase.

Reviewer #3 (Public review):

Summary:

The authors aimed to study the activation of gliogenesis and the role of newborn astrocytes in a post-ischemic scenario. Combining immunofluorescence, BrdU-tracing and genetic cellular labelling, they tracked the migration of newborn astrocytes (expressing Thbs4) and found that Thbs4-positive astrocytes modulate the extracellular matrix at the lesion border by synthesis but also degradation of hyaluronan. Their results point to a relevant function of SVZ newborn astrocytes in the modulation of the glial scar after brain ischemia. This work's major strength is the fact that it is tackling the function of SVZ newborn astrocytes, whose role is undisclosed so far.

Strengths:

The article is innovative, of good quality, and clearly written, with properly described Materials and Methods, data analysis and presentation. In general, the methods are designed properly to answer the main question of the authors, being a major strength.

Interpretation of the data is also in general well done, with results supporting the main conclusions of this article.

In this revised version, the points raised/weaknesses were clarified and discussed in the article.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Minor points:

- (1) Thanks for the clarification.*
- (2) Thanks for the clarification.*
- (3) The magnification is not apparent in Fig. 5.*

We had removed two brain slices (from 4 to 2) in order to increase the size of the image 2-fold. We have now further increased the TTC panel, 25% from the revised version, 125% from the original.

- (4) Thanks for the clarification.*
- (5) Thanks for the clarification.*
- (6) Thanks for the clarification.*
- (7) Thanks for the clarification.*
- (8) Thanks for the clarification.*

<https://doi.org/10.7554/eLife.96076.3.sa0>